

Chapter 7: Cell Proliferation & Regulation

- Why do cells divide?
- Before cells divide, they must replicate their DNA
- Mitosis = cell division
- How is mitosis regulated?
- What happens when regulation goes wrong?

Corresponds with OpenStax Biology 2e Chapter 10

Learning Objectives

Be able to

- describe mitosis process and
- describe how it is regulated
- describe difference between meiosis and mitosis
- describe cell cycle process



Why cells divide?

01 Growth

Cells divide to support growth, transforming a single fertilized egg into a multicellular organism. This process is fundamental to development and maturation.

02 Repair and Renewal

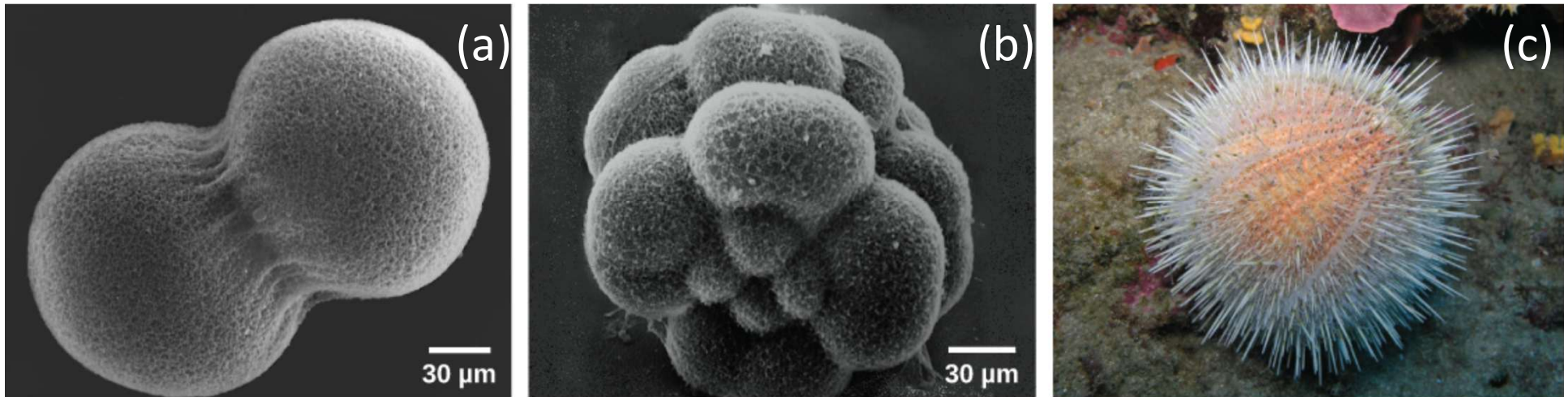
Essential for repairing damaged tissues and renewing cells in organs like skin and gut lining, ensuring the maintenance of healthy tissues.

03 Reproduction

In single-celled organisms, cell division is the primary mode of asexual reproduction, allowing them to multiply and propagate.

04

Visual Example



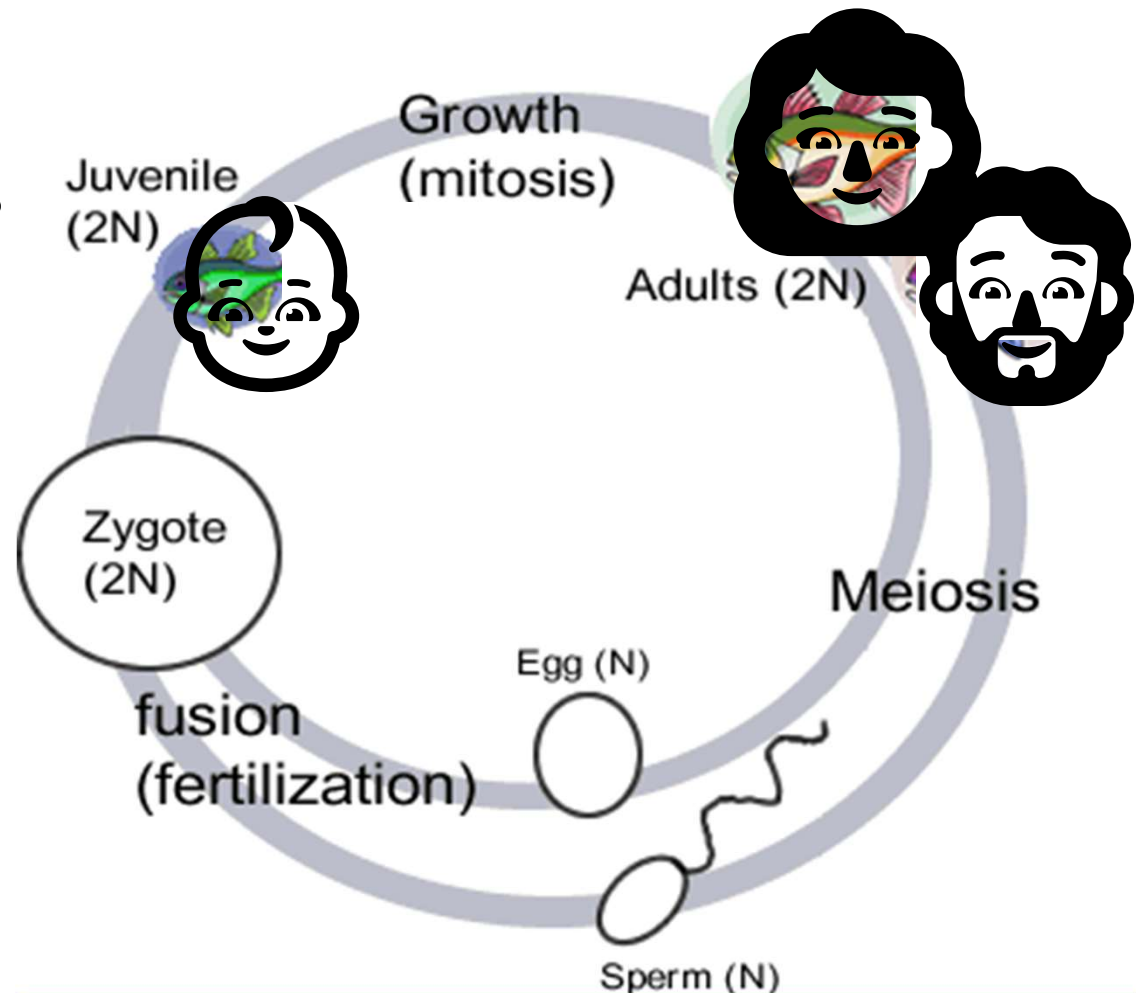
A sea urchin begins life as (a) single diploid cell (zygote) divides to form two genetically identical daughter cells (SEM). (b) 16 cells after four rounds of cell division. (c) After many rounds of cell division, the individual develops into a complex, multicellular organism, i.e., mature sea urchin

Why do cells divide?

There are 2 types of cell division

1. Meiosis

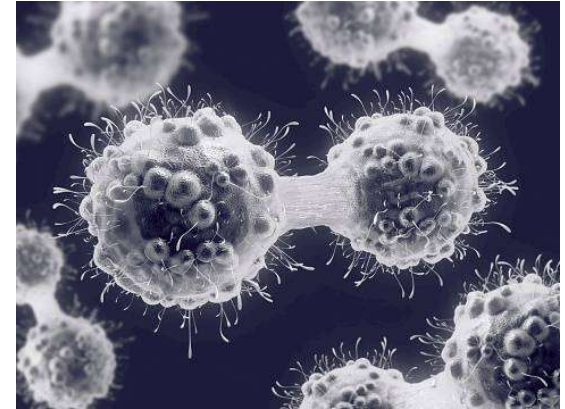
- makes gametes (egg & sperm) which are not identical
- only for **sexual** reproduction



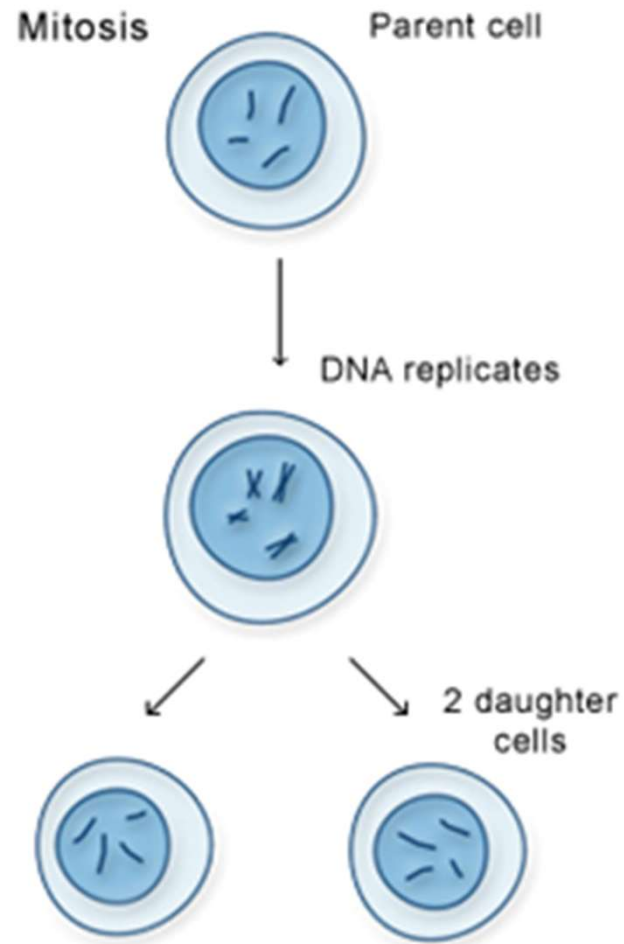
Why do cells divide?

There are 2 types of cell division

- 2. **Mitosis** makes identical cell copies
 - process by which:
 - organisms grow
 - *Remember: to grow we don't get bigger cells, we get more cells*
 - old cells can be replaced
 - some organisms can **asexually** reproduce
 - some organisms can regenerate body parts

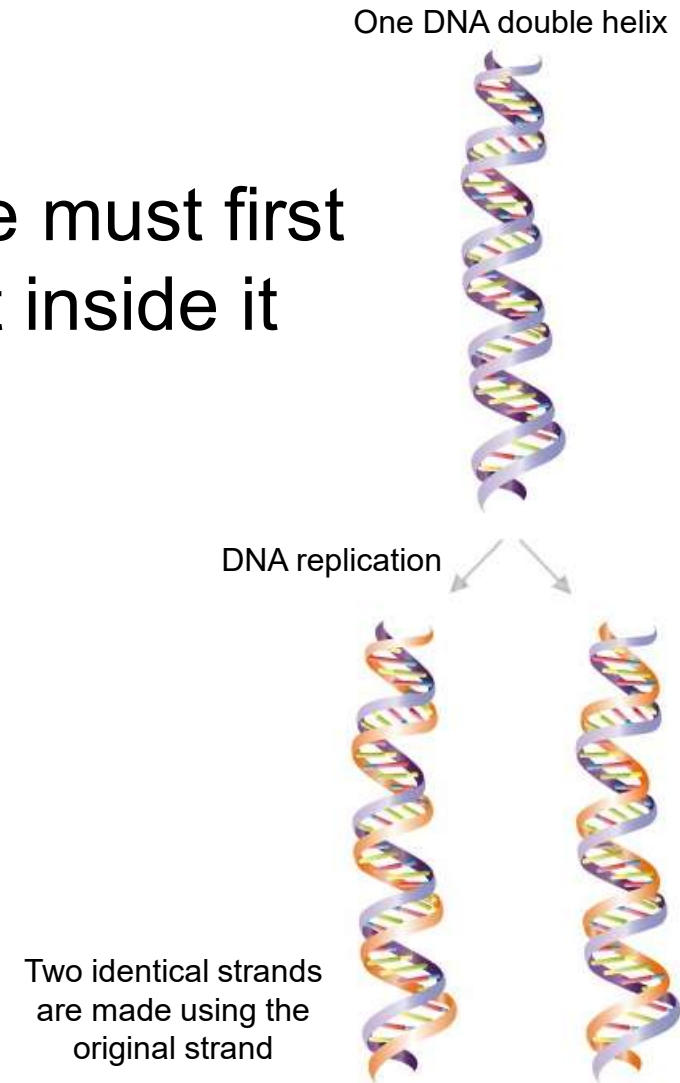


Mitosis



Before cells divide, they must replicate their DNA

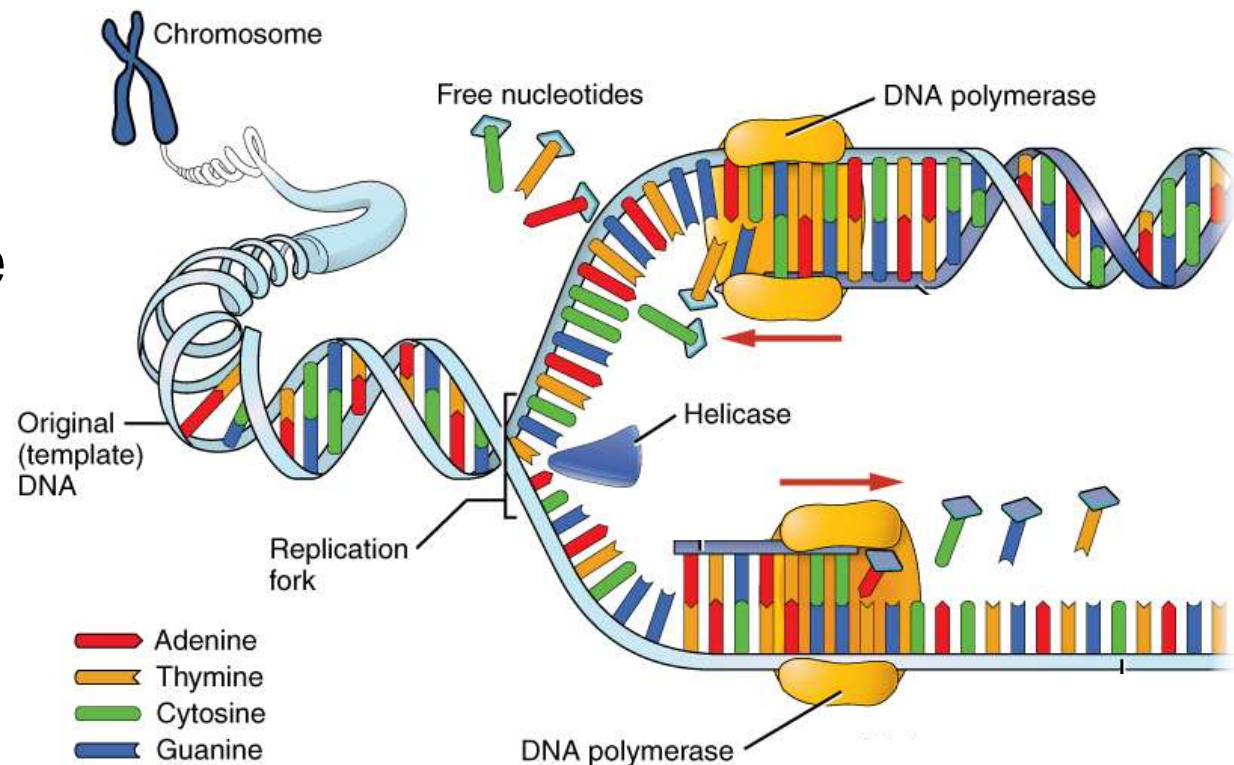
- *Remember:* all cells have DNA
 - In order to make a new cell, we must first make a copy of our DNA to put inside it
 - So before any type of cell division (mitosis or meiosis) cells must duplicate all of their DNA
 - = **DNA replication**



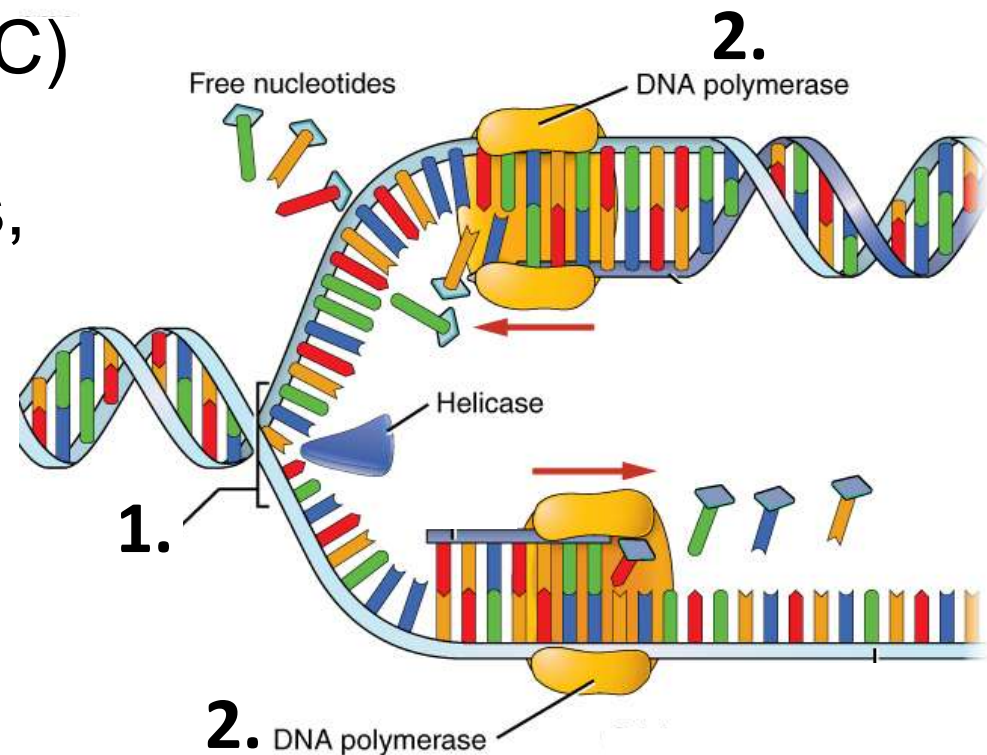
- DNA replication requires 3 ingredients

- the original DNA double helix
- free nucleotides (A, T, G, C)
- several enzymes

- e.g. DNA helicase & DNA polymerase

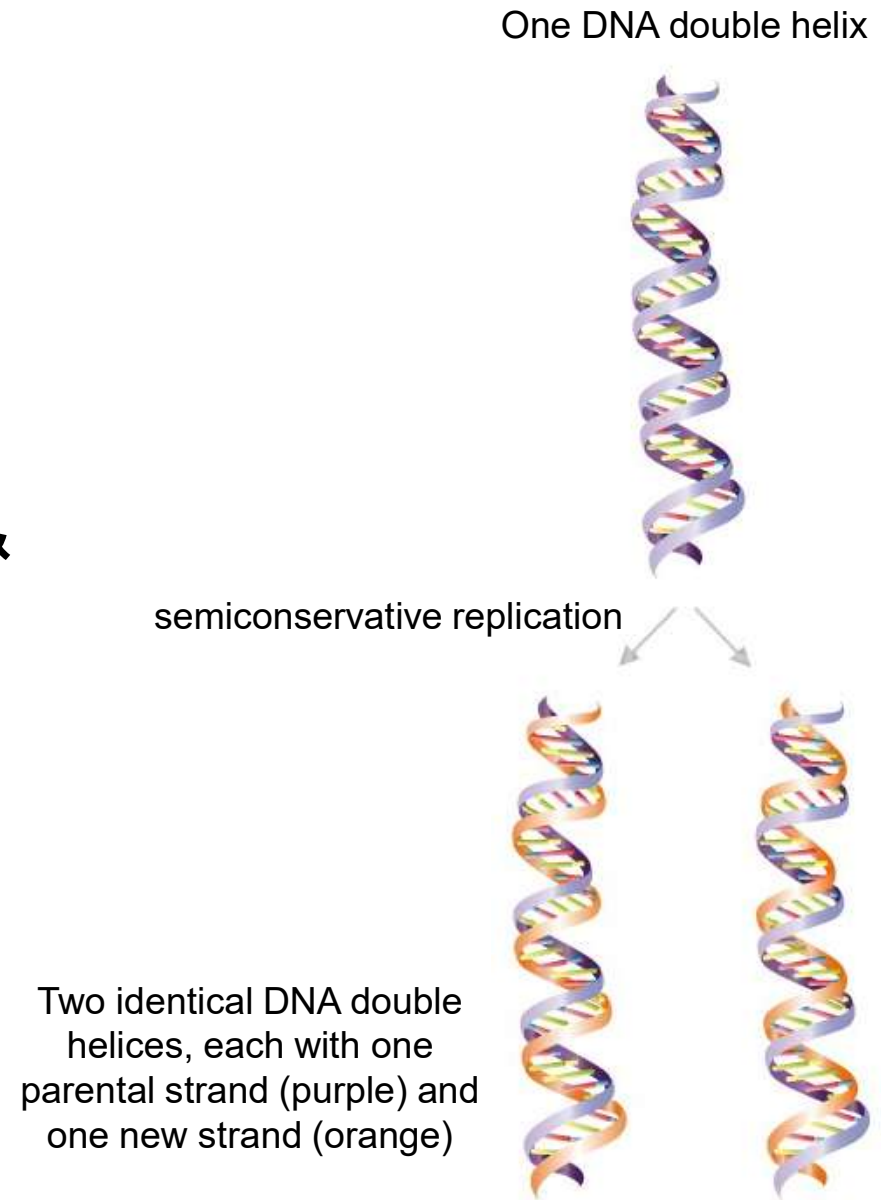


- DNA replication happens in 2 main steps
 - 1. the original double helix is pulled apart (unzipped) by the enzyme **DNA helicase**
 - 2. the enzyme **DNA polymerase** pairs free nucleotides (A, T, G, C) with each of the now single-stranded sides, building new double-stranded DNA



- DNA replication

- When replication is complete, each of the final DNA copies have one old original strand & one newly made strand = **semiconservative replication**

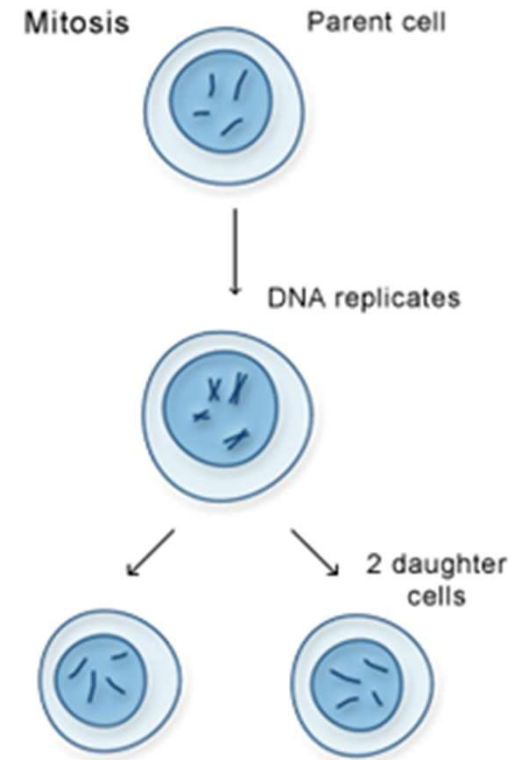


Mitosis = cell division

- Now that there are 2 copies of DNA, the cell can divide in half, placing 1 copy in each cell

– **Mitosis** (also called mitotic cell division) = process by which 1 parent cell gives rise to 2 identical daughter cells

- *Cells that undergo mitosis follow a cell cycle*



U.S. National Library of Medicine

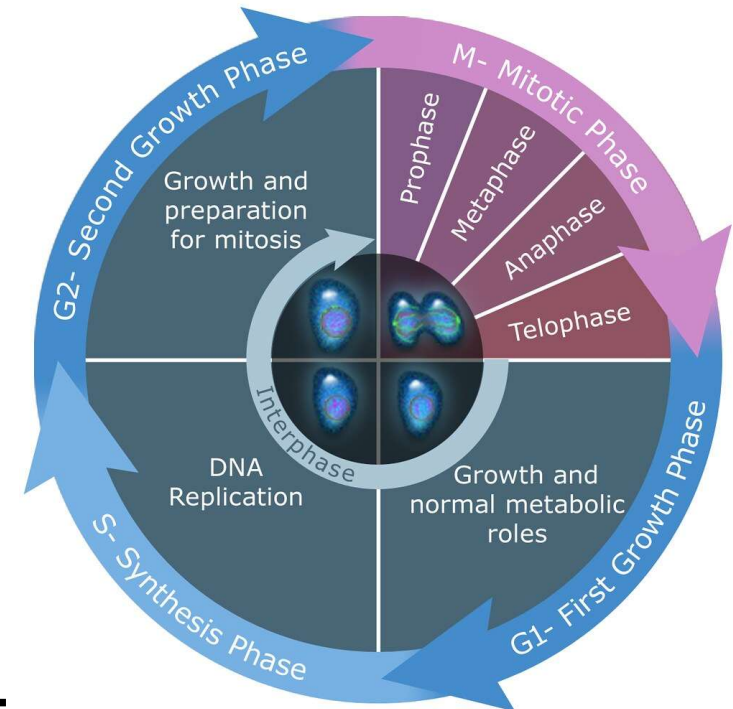
- Cells that undergo mitosis go through a repeating pattern: **cell cycle**

- Divide
- Grow
- (in some cases) Differentiate
- Divide again, etc...

- some cells **differentiate** (only once): they become specialized cells for specific functions

–e.g. skin cells, or heart cells, or liver cells

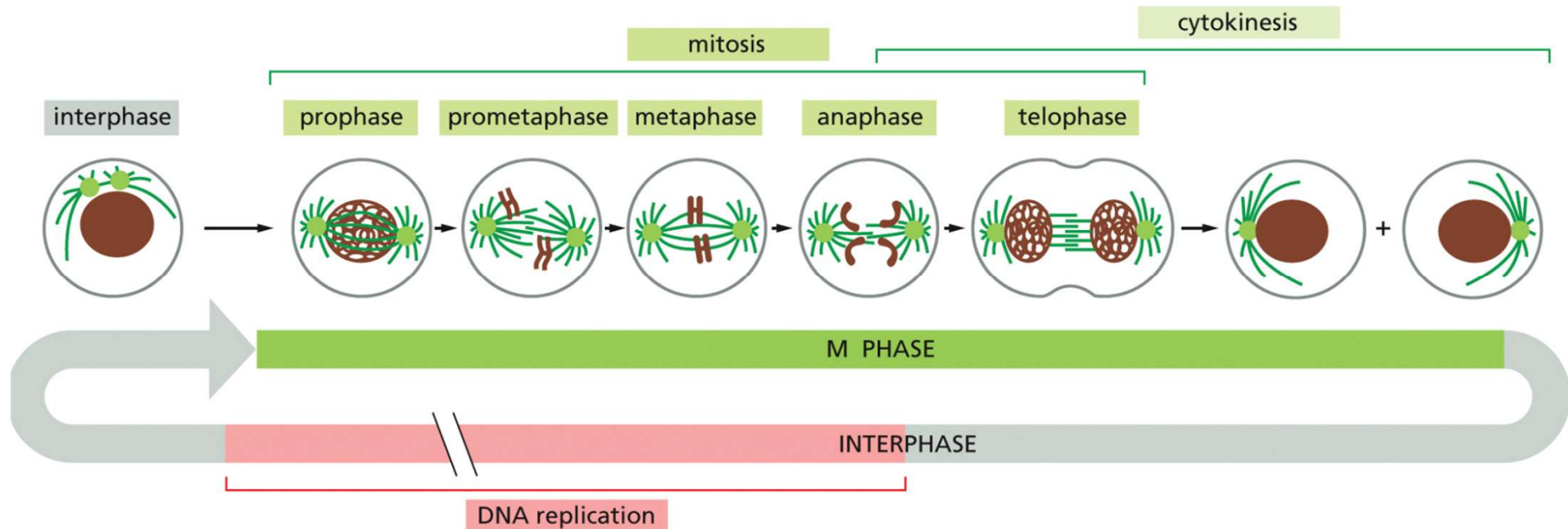
- *BUT: not all cells divide*



Growth phases are all in blue, divide phases in pink

- **Organisms have cells that can & can NOT divide**
 - Stem cells & most other cells in animals can divide: they follow the cell cycle
 - Stem cells have not differentiated yet – can eventually differentiate into a variety of cell types
 - Regular cells have differentiated but keep dividing: liver cells produce more liver cells, etc.
 - Certain animal cells have differentiated & will never divide again: leave the cell cycle at the earliest growth phase
 - e.g. most heart & brain cells can NOT divide
-

■ Cell cycle



Cycle Definition

Cell cycle is an ordered sequence of events from one division to the next. It consists of interphase and the mitotic (M) phase.

Cycle Phases

Interphase includes G_1 , S , and G_2 phases, focusing on cell growth and DNA replication. The M phase involves mitosis and cytokinesis, culminating in cell division.

■ Cell cycle

- Life of a cell formation to division
- Hours to years
- Bacteria-20 min
- Paramecium-6 hours
- Onion root tip -20 hr
- Human cell-90 hr

Paramecium



0.06-0.3mm

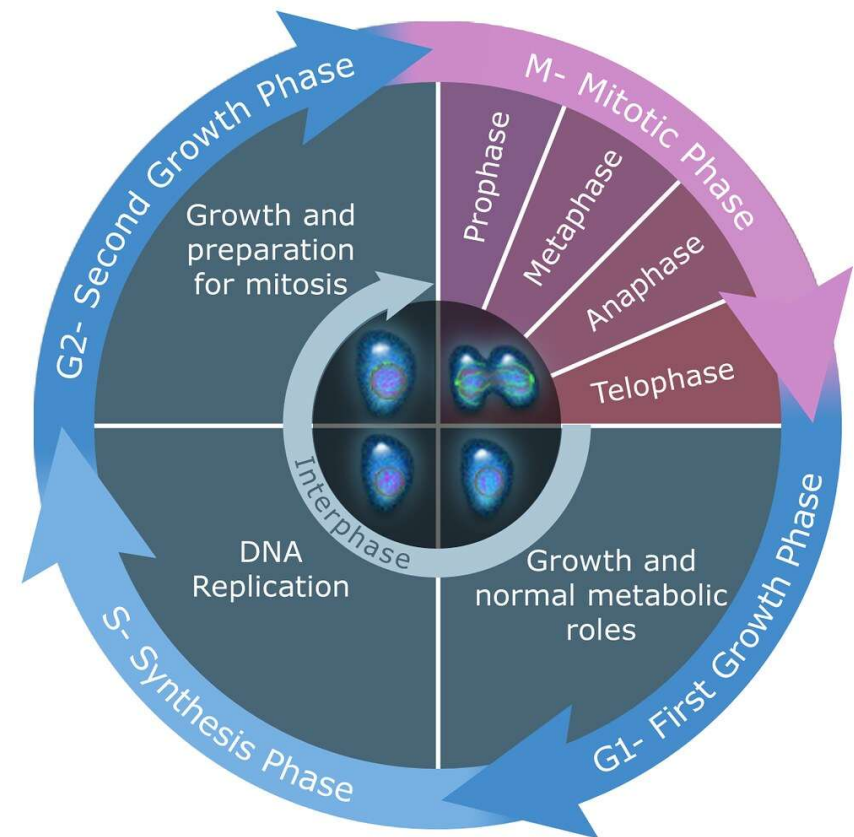
Mitosis = cell division



- Cells that divide through mitosis follow the cell cycle

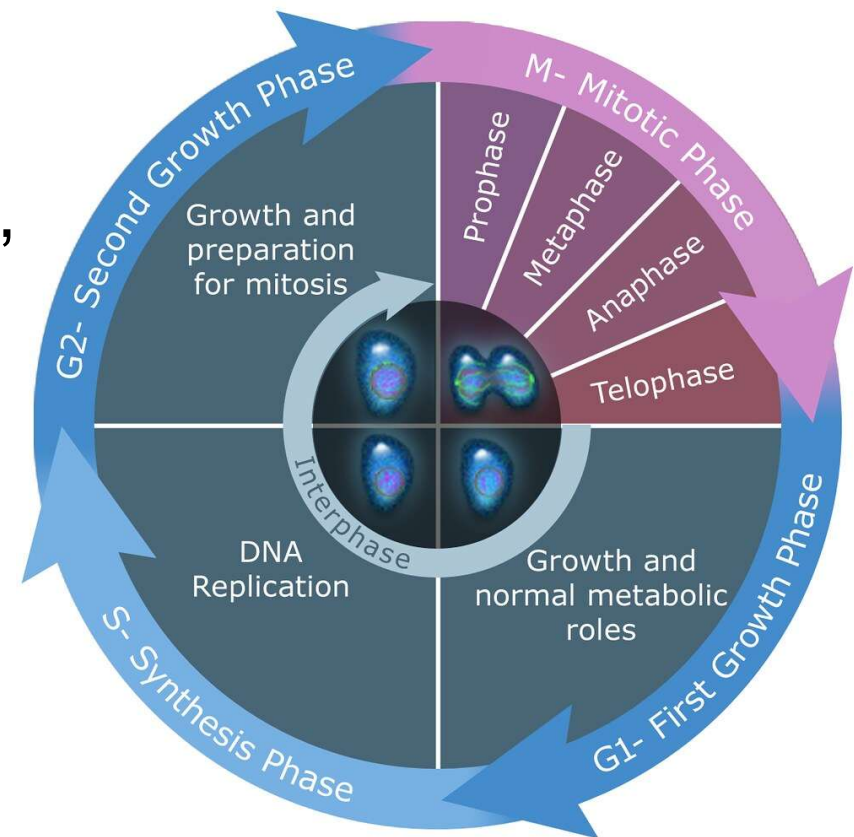
- **Interphase** = a time for acquisition of nutrients, growth, & DNA replication

- Cells spend most of their time in interphase
 - (*blue in figure →*)
 - Includes G1, S, & G2 phases



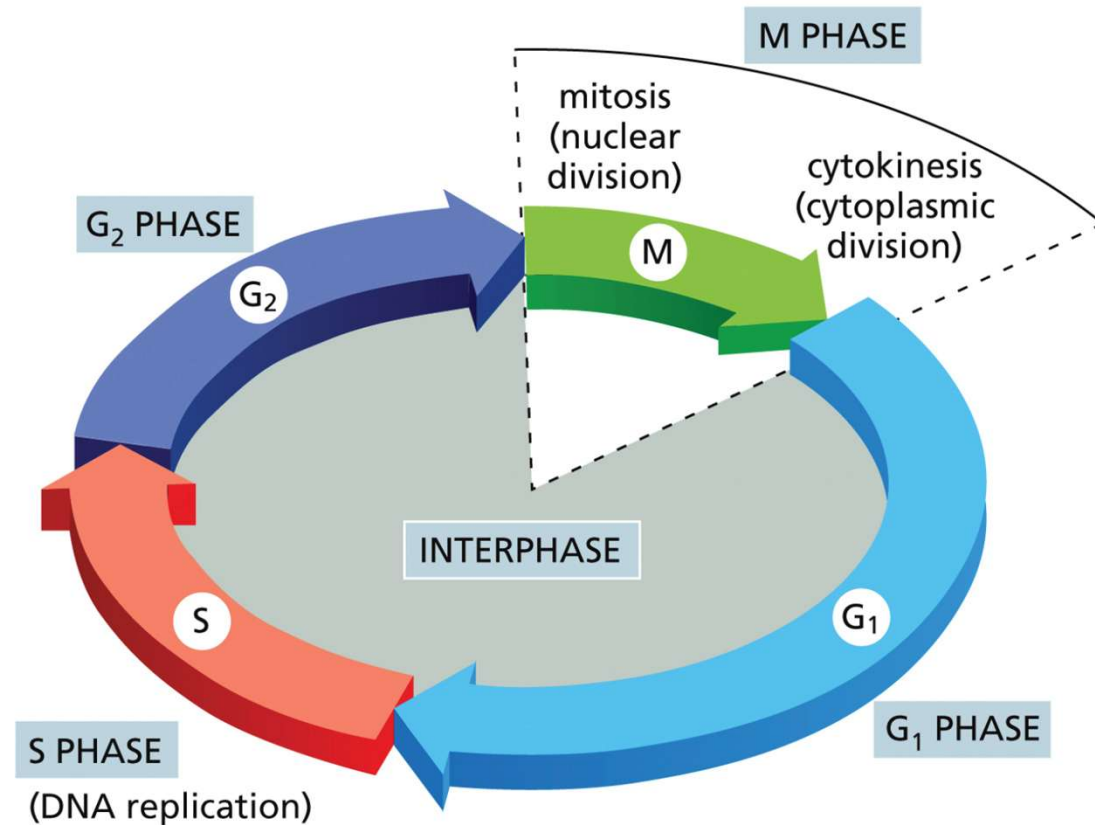
- Interphase is divided into 3 phases

- **G₁** (growth phase 1) = acquisition of nutrients & growth to proper size
- **S** (synthesis phase) = DNA replication: all DNA is copied, or “synthesized”
- **G₂** (growth phase 2) = completion of cell growth & final preparation for mitosis



G₁ Phase

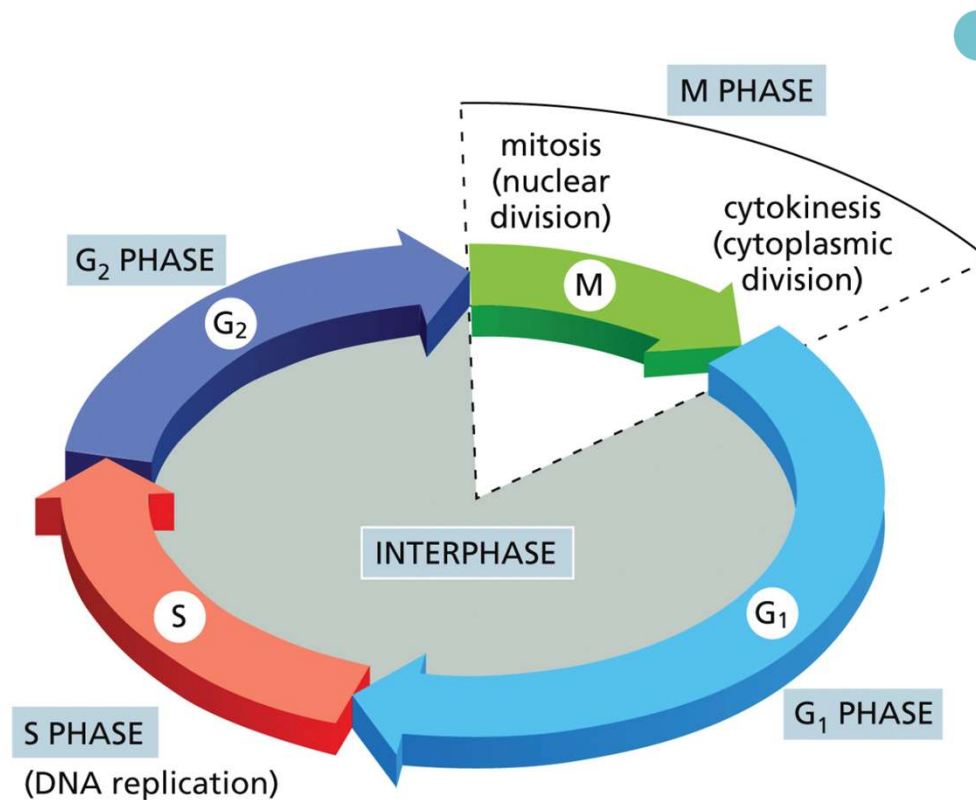
The G₁ phase is the primary primary period of cell growth. growth. It ends with a critical critical checkpoint that determines whether the cell cell will divide, delay, or enter enter a non-dividing state (G₀).



G₁ Phase

- first gap phase
- LONGEST phase
- Cell grows physically larger
- copies organelles(MC,ER,RIB,LYSO)
- Synthesis of protein,RNA,No DNA
- Nucleolus produces r DNA. mRNA, tRNA are synthesized.
- Metabolic rate high
- G1 cyclins rise-signal-preapre-replication

S Phase Synthesis



● S Phase

During the S phase, DNA replication occurs, creating sister chromatids. The centrosome also duplicates, preparing for mitosis.

● Replication Process

DNA replication is a highly regulated process ensuring that each chromosome is copied accurately, maintaining genetic stability.

S Phase Synthesis

- cell synthesizes DNA in its nucleus.
- Histones are synthesized
- Each chromosome divided into 2 chromatids
- It also duplicates centrosome.
- The centrosomes help separate DNA during M phase.

G₂ Phase

G₂ Phase

The G₂ phase is the final preparation stage before mitosis. The cell continues to grow and synthesizes proteins necessary for division.

G₂ Checkpoint

The G₂ checkpoint ensures ensures that DNA replication is complete and and without damage. This This prevents cells with with flawed DNA from entering mitosis.

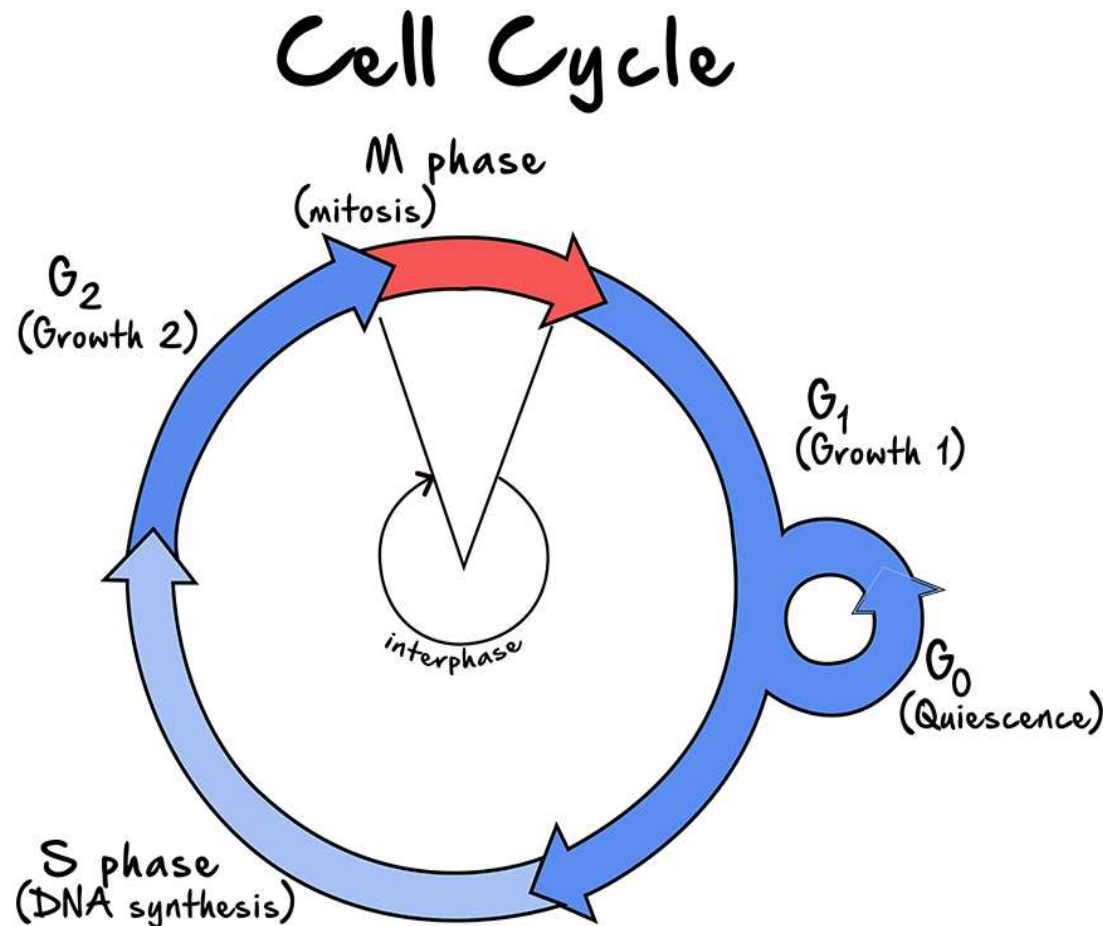
Quality Control

Quality control mechanisms in G₂ ensure ensure that all chromosomes are properly duplicated and and ready for segregation.

G₂ Phase

- second gap phase
- Nucleus increase
- cell grows more, makes proteins & organelles
- ends when mitosis begins.
- Followed by mitotic phase

G₀ Phase Quiescence



The G₀ phase is a non-dividing state where cells can enter temporarily or permanently. Examples include neurons, muscle cells, and liver cells.

- Cells that divide through mitosis follow the cell cycle

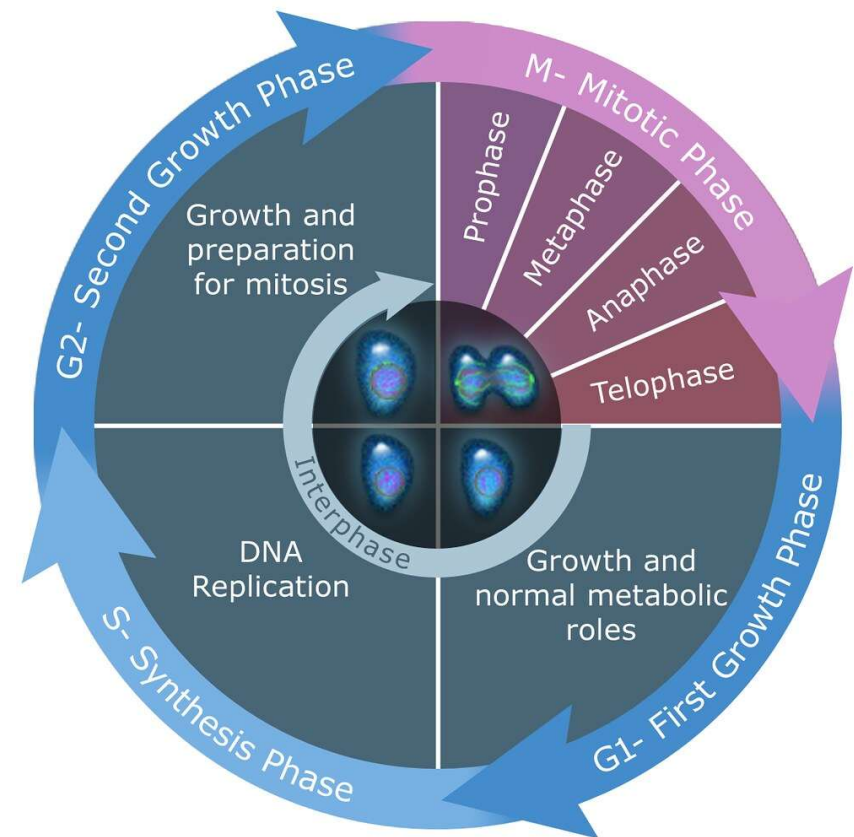
- **M-phase** = mitotic phase + cytokinesis

- Cells spend time to divide

- Nucleus and cytoplasm

- (*pink in figure →*)

- Includes Prophase, Metaphase, Anaphase, Telophase, Cytokinesis



- **Mitosis** = actual cell division: 1 copy of DNA & half of cytoplasm goes into each of 2 daughter cells
 - Duplicated DNA/chromosomes are separated in half & pulled to opposite ends of the cell
 - Then the cell divides along the middle
 - *has several phases*
 - *More to come*

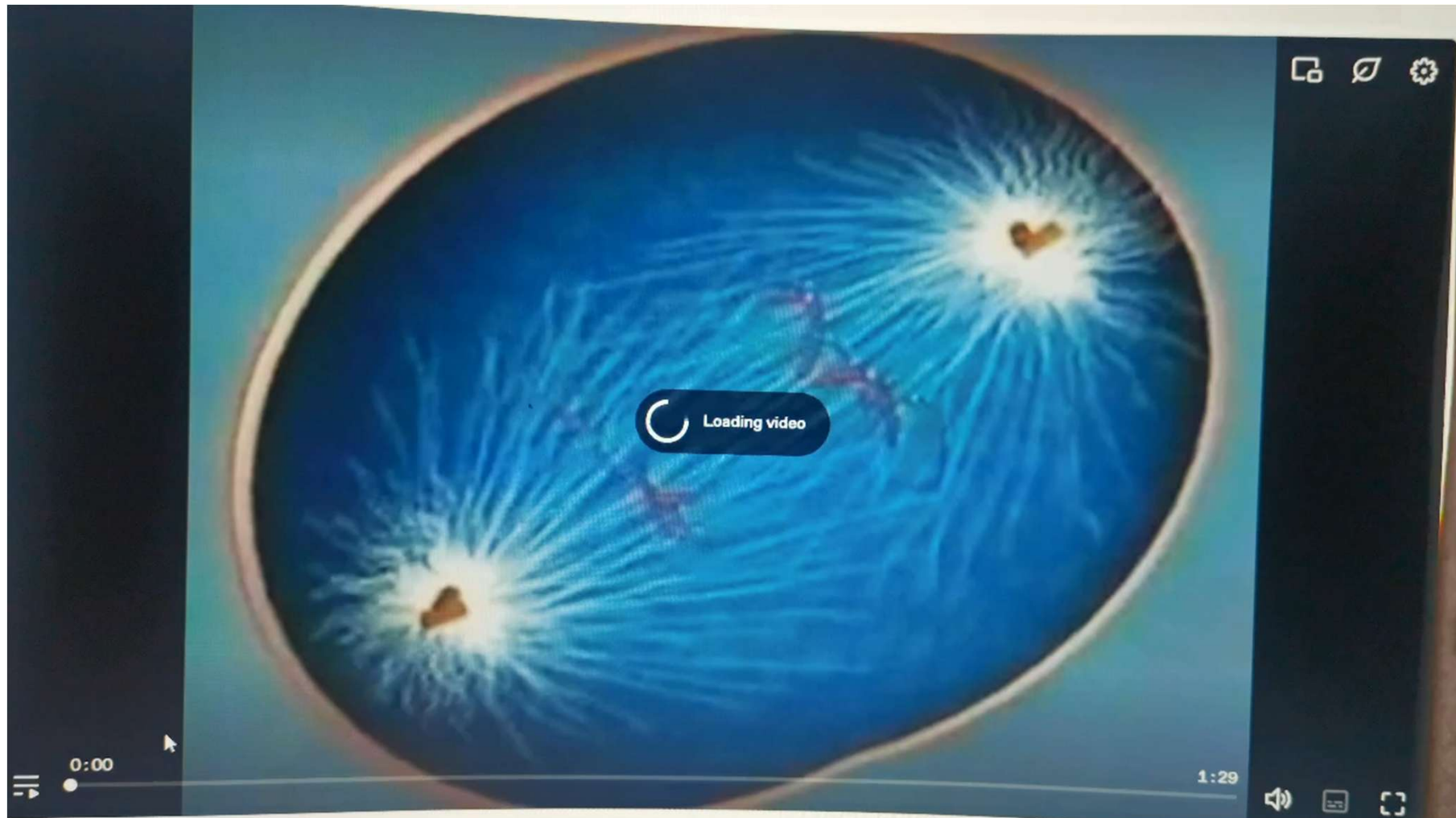
Two copies of DNA being pulled to either side in a cell about to divide down the middle to make two cells

DNA inside nucleus of a cell not dividing



■ Mitosis

General reading



Prophase Events

Chromatin Condensation

Condensation

During prophase, chromatin condenses into visible chromosomes, making them easier to segregate during division.

Centrosome Migration

Migration

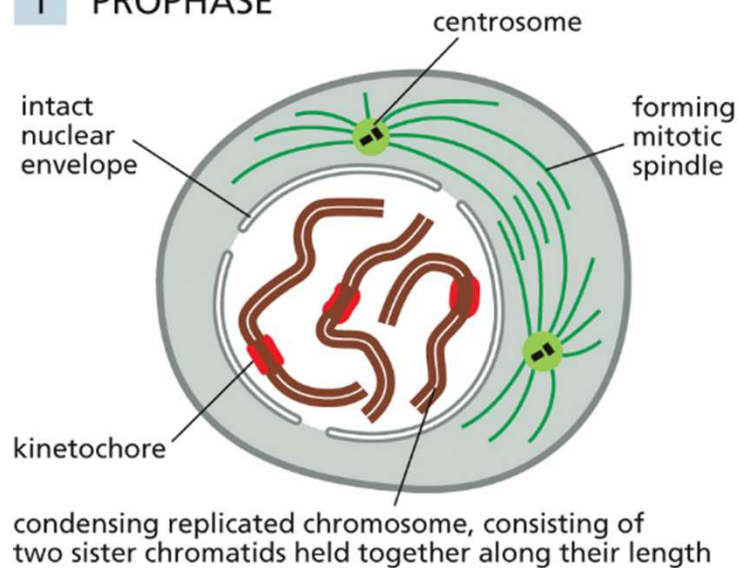
The centrosomes migrate to opposite poles of the cell, initiating the formation of the mitotic spindle.

Nucleolar Disappearance

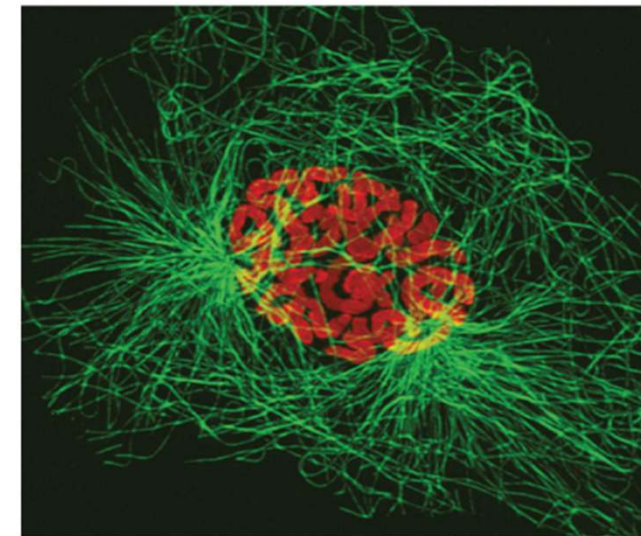
Disappearance

The nucleolus disappears as the cell prepares for nuclear envelope breakdown and chromosome segregation.

1 PROPHASE



At **prophase**, the replicated chromosomes, each consisting of two closely associated sister chromatids, condense. Outside the nucleus, the mitotic spindle assembles between the two centrosomes, which have replicated and moved apart. For simplicity, only three chromosomes are shown. In diploid cells, there would be two copies of each chromosome present. In the fluorescence micrograph, chromosomes are stained **orange** and microtubules are **green**.



Micrograph courtesy of Julie Canman and Ted Salmon

10 μm

Prometaphase Attachments

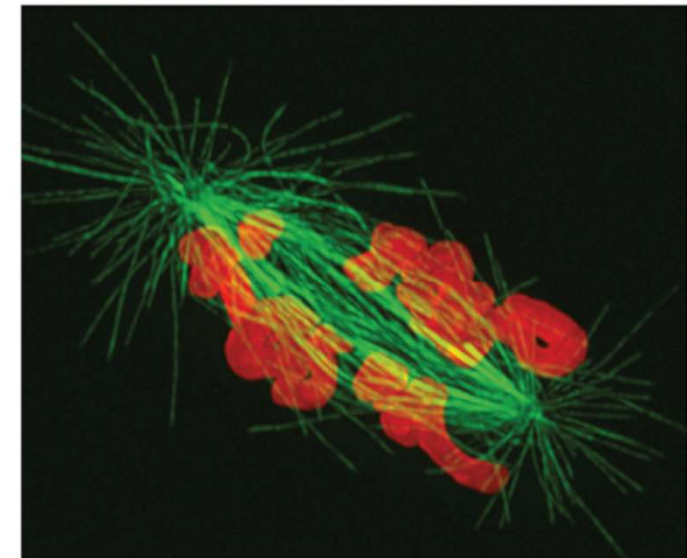
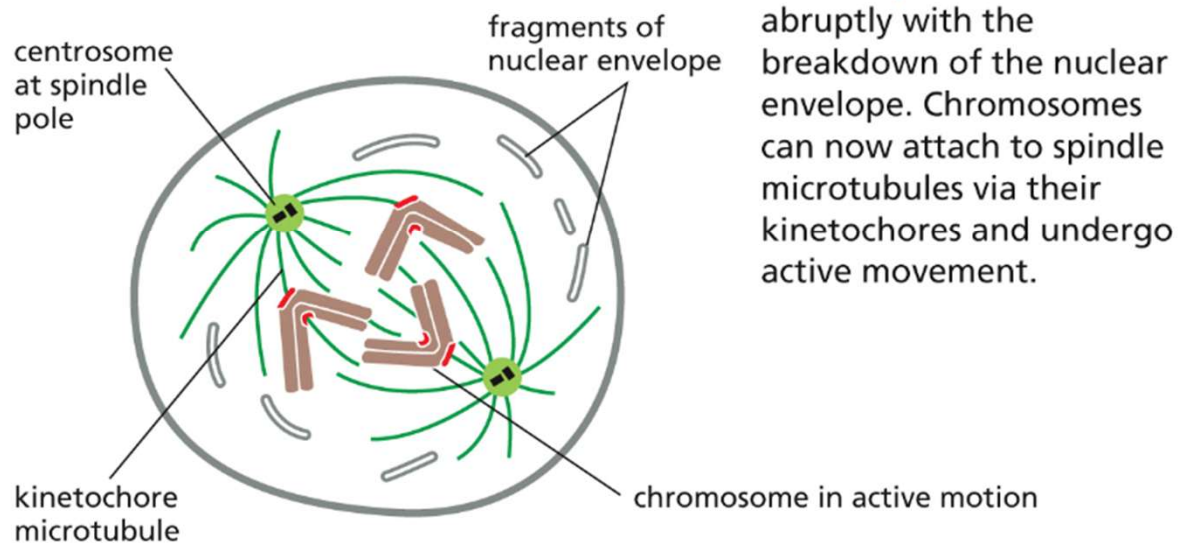
Nuclear Envelope

In prometaphase, the nuclear envelope disassembles, allowing spindle microtubules to invade the nuclear area.

Kinetochores Formation

Kinetochores formation at the centromeres enables microtubules to attach and align chromosomes correctly.

2 PROMETAPHASE



Micrograph courtesy of Julie Canman and Ted Salmon

Metaphase Alignment

Chromosome Alignment Alignment

During metaphase, chromosomes align at the metaphase plate, ensuring equal segregation of genetic material.

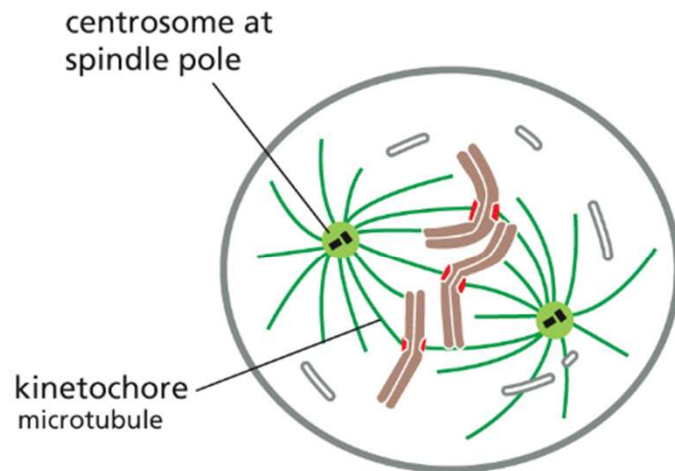
Kinetochores Attachment

Kinetochores microtubules attach to sister kinetochores, pulling them toward opposite poles to maintain tension.

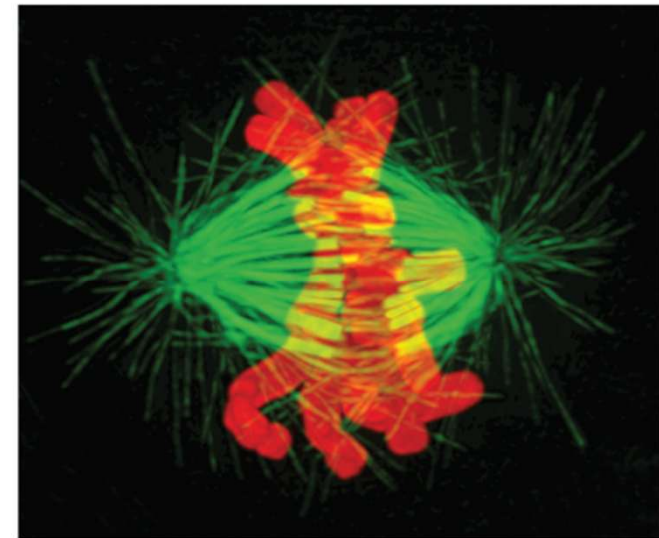
Spindle Assembly Checkpoint

The spindle assembly checkpoint ensures that all chromosomes are properly attached before anaphase begins.

3 METAPHASE



At **metaphase**, the chromosomes are aligned at the equator of the spindle, midway between the spindle poles. The kinetochore microtubules attach sister chromatids to opposite poles of the spindle.



Micrograph courtesy of Julie Canman and Ted Salmon

Anaphase Separation

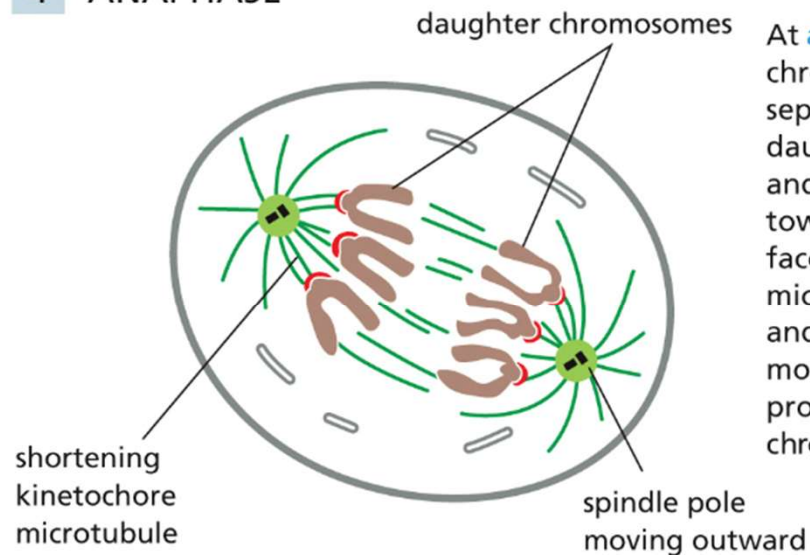
Sister Chromatid Separation

Anaphase begins with the separation of sister chromatids, which are now considered individual chromosomes.

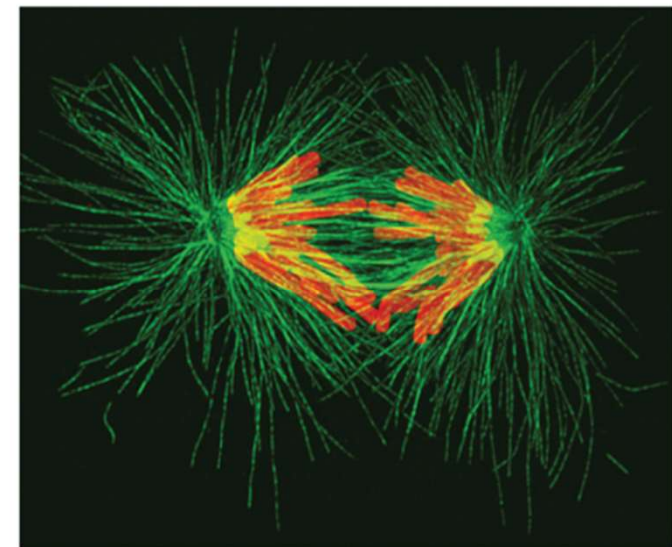
Chromosome Movement

Motor proteins transport chromosomes toward opposite poles, ensuring accurate segregation.

4 ANAPHASE



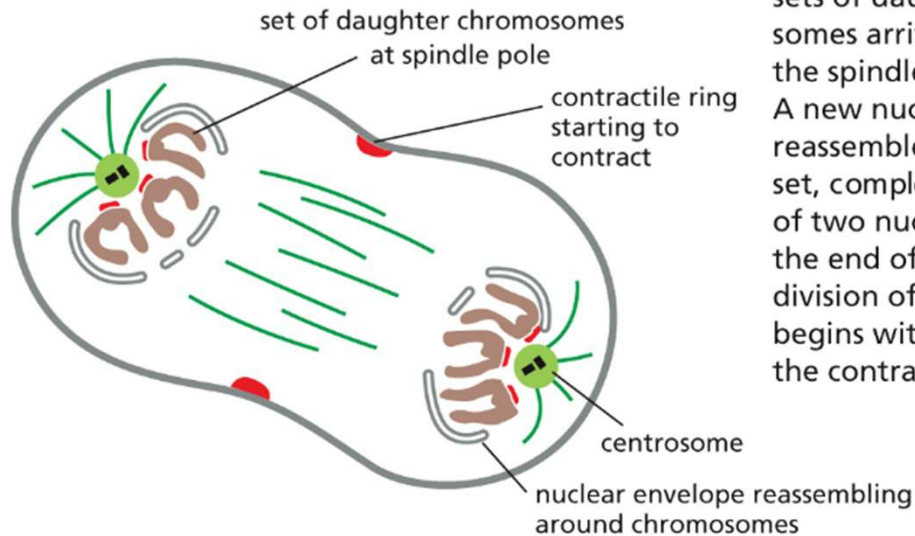
At **anaphase**, the sister chromatids synchronously separate to form two daughter chromosomes, and each is pulled slowly toward the spindle pole it faces. The kinetochore microtubules get shorter, and the spindle poles also move apart; both processes contribute to chromosome segregation.



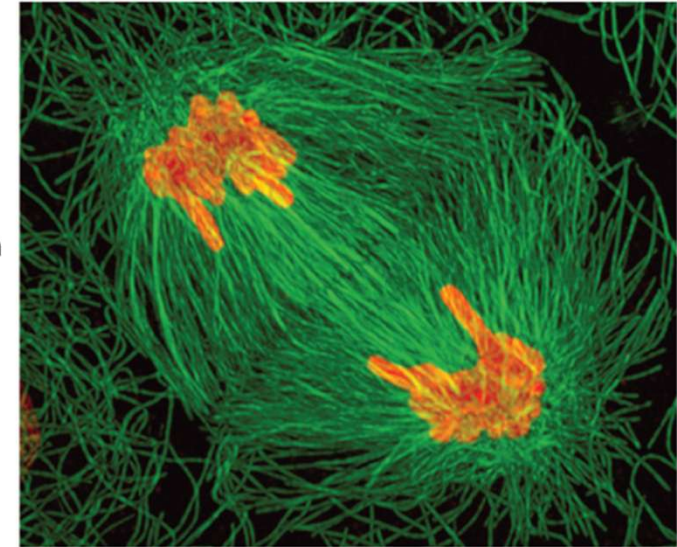
Micrograph courtesy of Julie Canman and Ted Salmon

Telophase Reformation

5 TELOPHASE



During **telophase**, the two sets of daughter chromosomes arrive at the poles of the spindle and decondense. A new nuclear envelope reassembles around each set, completing the formation of two nuclei and marking the end of mitosis. The division of the cytoplasm begins with contraction of the contractile ring.



Micrograph courtesy of Julie Canman and Ted Salmon

Nucleolar Reappearance

The nucleoli reappear, completing the reassembly of nuclear structures.

Nuclear Envelope Reformation

In telophase, nuclear envelopes re-form around the separated chromosome sets, creating two distinct nuclei.

Chromosome Decondensation

Chromosomes decondense back into chromatin, preparing for interphase.

Cytokinesis – Division of cytoplasm

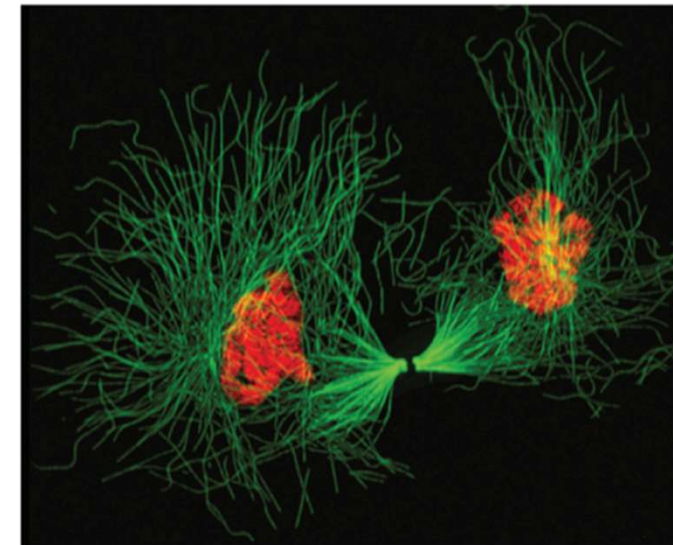
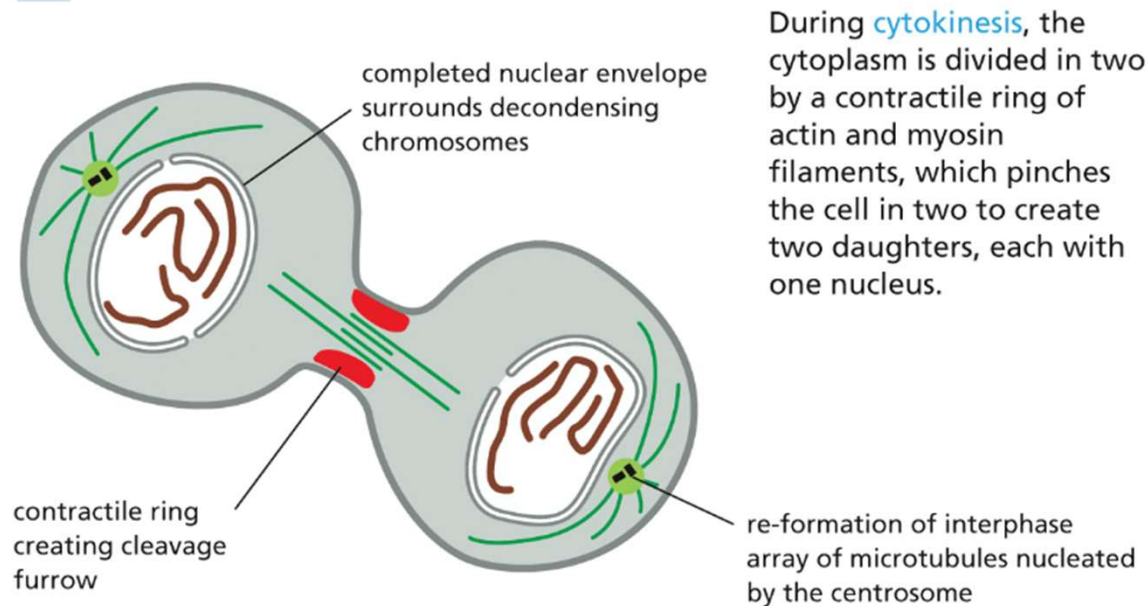
Cytokinesis Definition

Cytokinesis is the process of cytoplasmic division that usually overlaps with late mitosis, resulting in two separate daughter cells.

Mechanisms

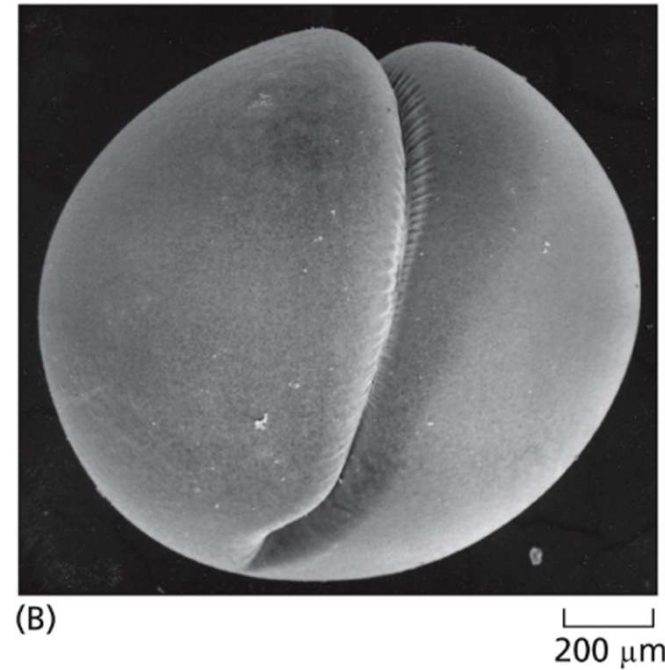
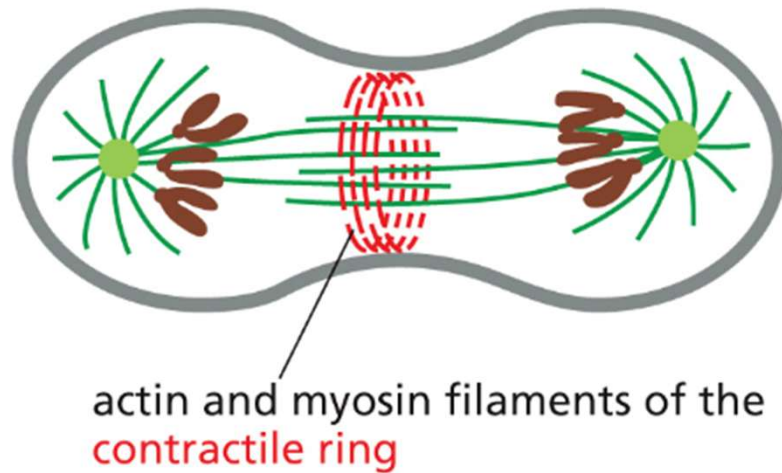
The mechanisms of cytokinesis differ between animal and plant cells, yet both achieve accurate partitioning of cellular contents.

6 CYTOKINESIS



Micrograph courtesy of Julie Canman and Ted Salmon

Cleavage



Contractile Ring Formation

In animal cells, a contractile ring of actin and myosin forms beneath the plasma membrane.

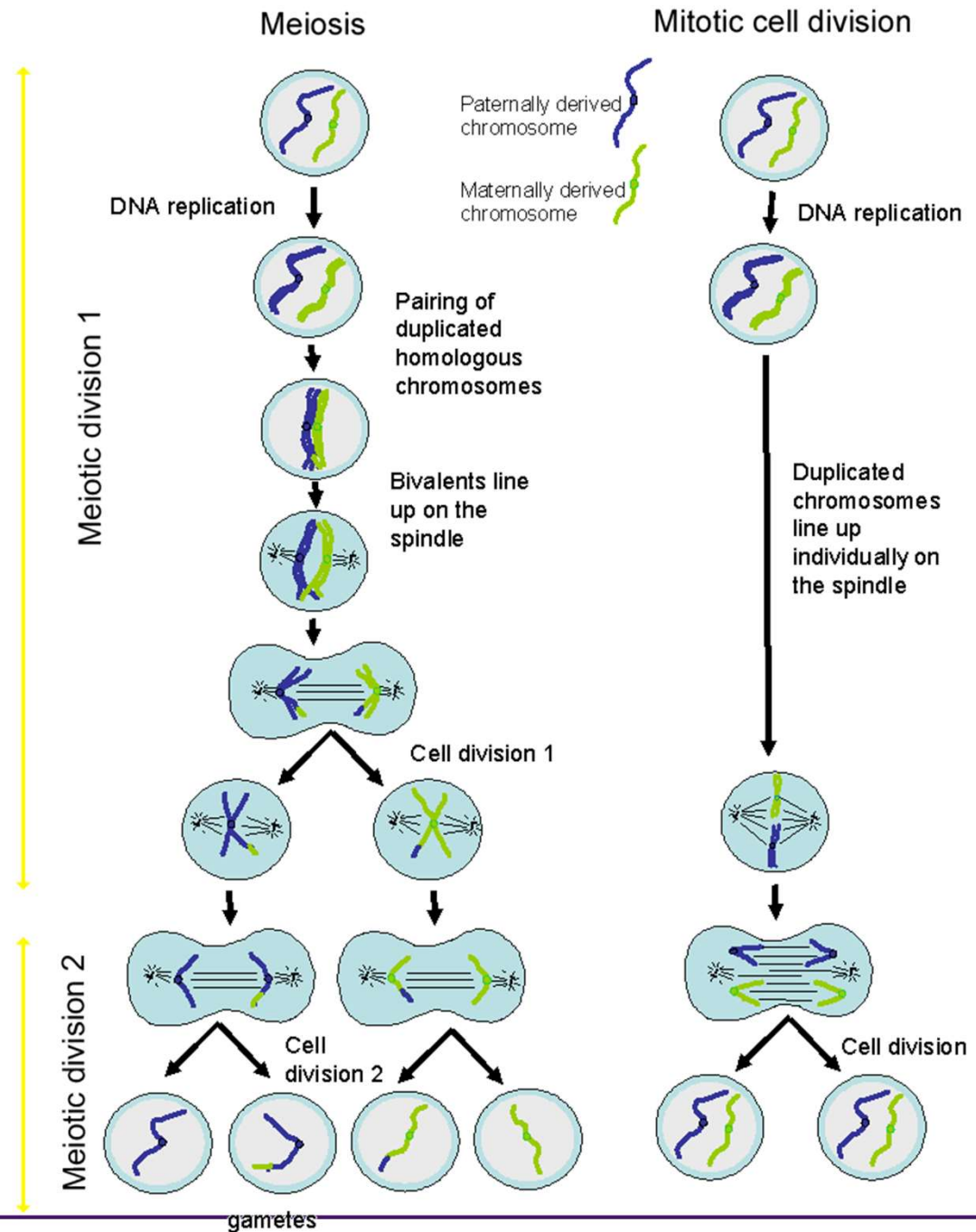
Cleavage Furrow

The contractile ring contracts, forming a cleavage furrow that pinches the cell into two daughter cells.

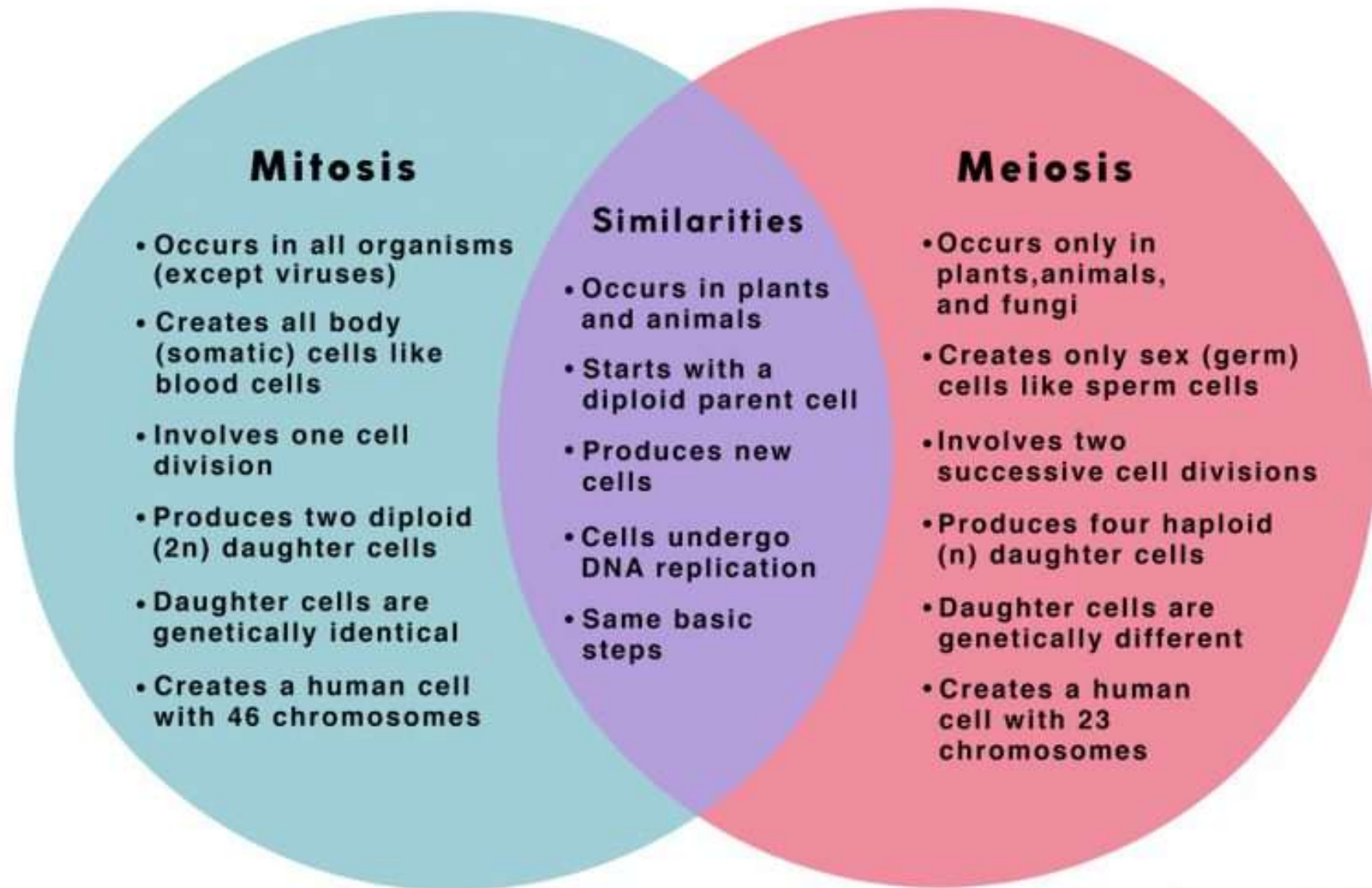
Completion

This process ensures accurate partitioning of cytoplasm, organelles, and plasma membrane between the two new cells.

Difference between meiosis and mitosis



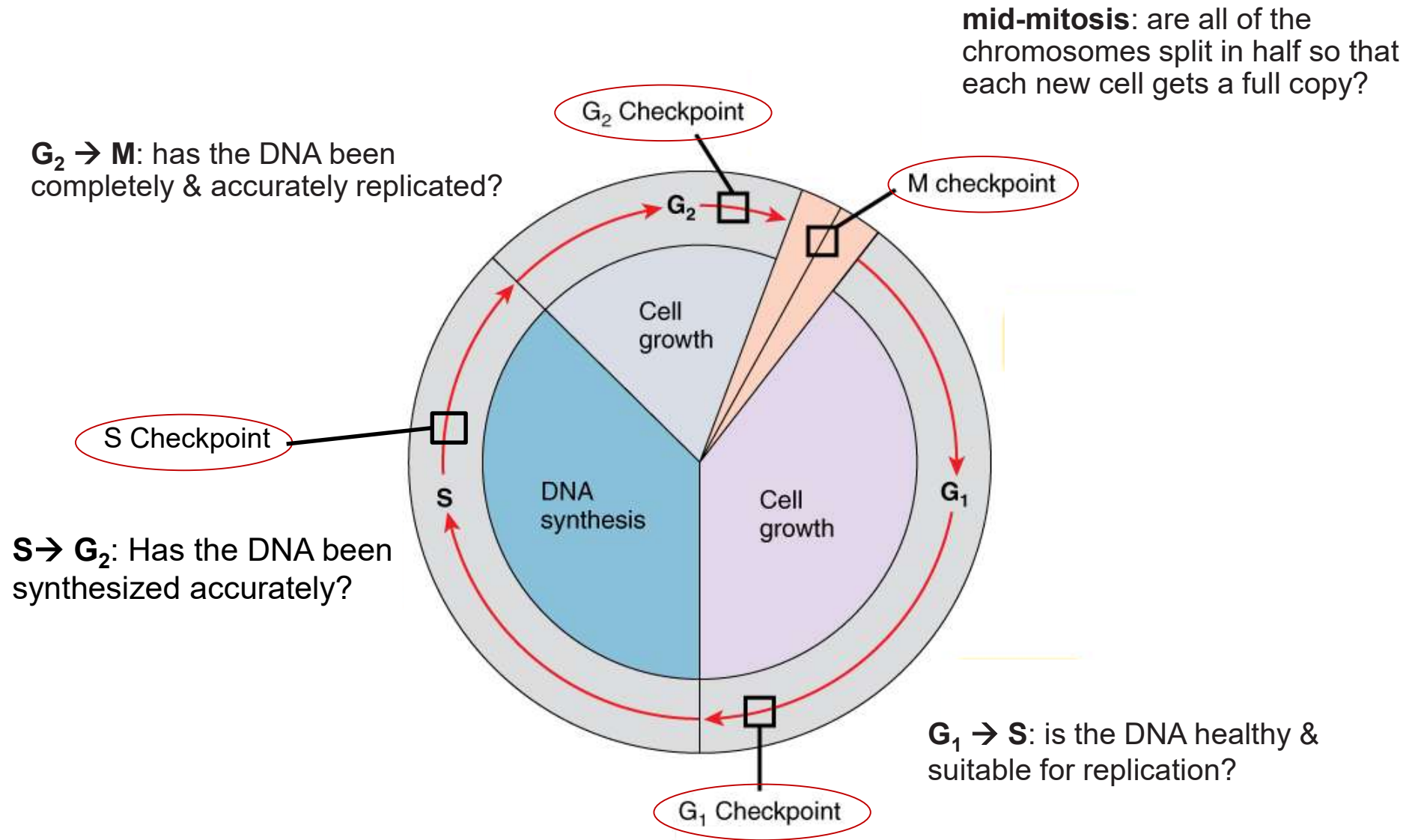
Mitosis and Meiosis Venn Diagram



How is mitosis regulated?

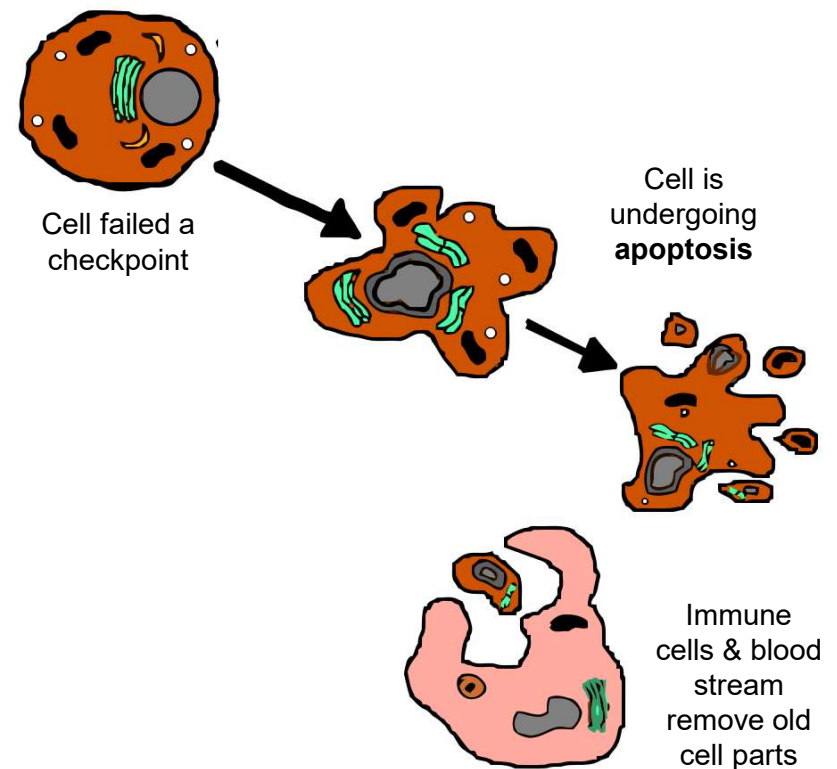
- The cell cycle is regulated by several checkpoints
 - **Checkpoints** = enzymes (proteins) that move through the cell to make sure it is healthy before it divides
 - 3 big checkpoints check to be sure:
 - the DNA is healthy & suitable for replication
 - DNA has been fully replicated correctly
 - DNA/chromosomes have split in half perfectly so that each cell will have a full copy
-

Cell cycle checkpoints:



- What happens if a cell fails a checkpoint?
 - **Apoptosis** = programmed cell death (cell suicide): it seems extreme that a cell would kill itself, but it is very useful for 3 big reasons

- 1. destroys cells that are not in good condition (infected with a virus, has mutations, etc.)

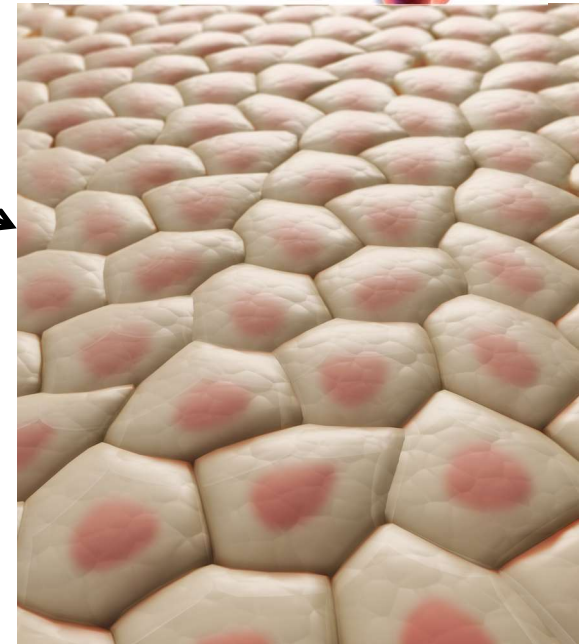


– **2.** removes old cells so that new ones can take their place

- red blood cells only live —————→
~120 days before undergoing
apoptosis & being replaced



- skin cells only live ~20 days ↘
- colon cells only live for ~4 days
before being replaced!



- 3. removes unneeded cells during development
 - e.g. hands & feet start out paddle-shaped in early embryos – apoptosis removes unneeded cells, allowing for individual fingers & toes

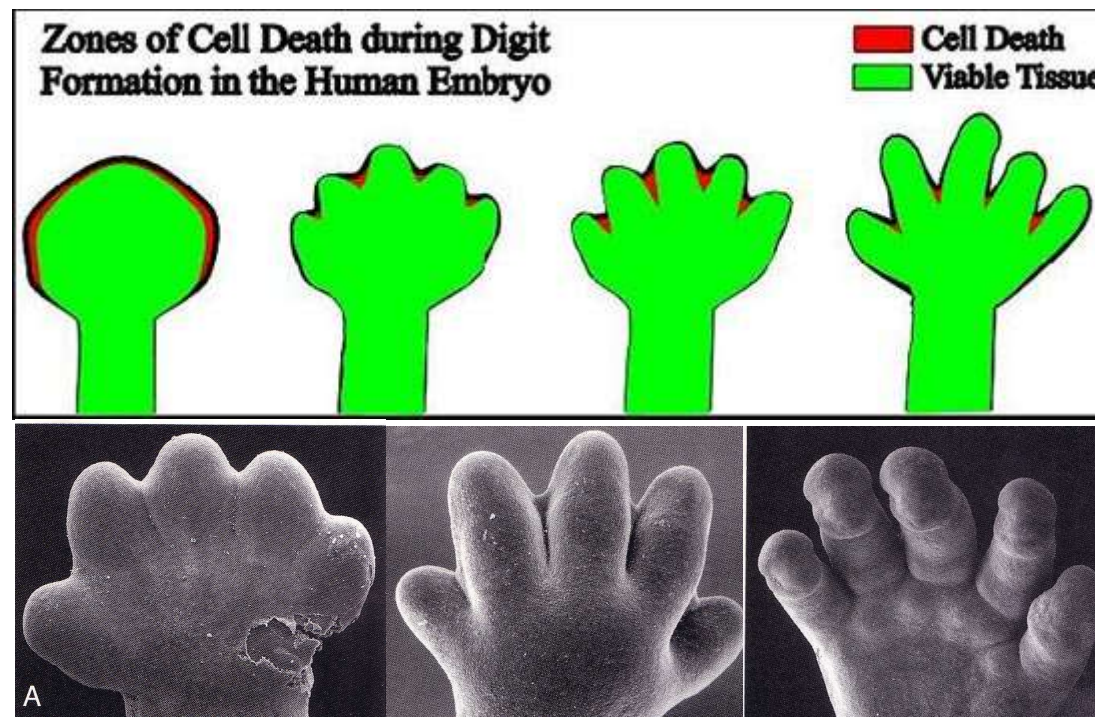


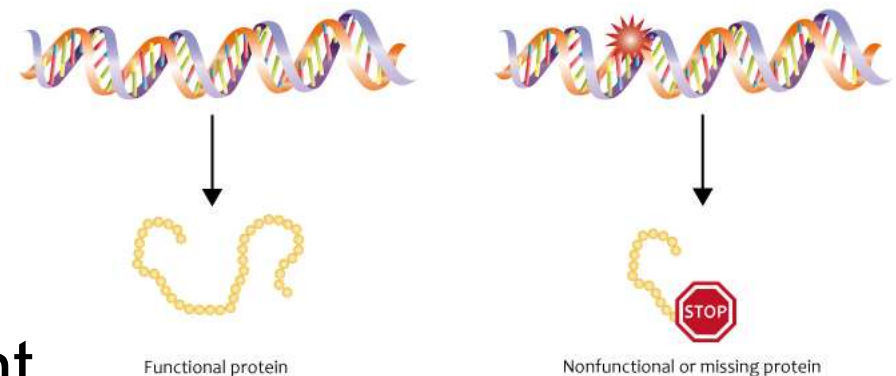
Figure: <https://www.slideserve.com/xanto/development-of-the-limbs>

Figure: <https://doctorlib.info/surgery/plastic/89.html>

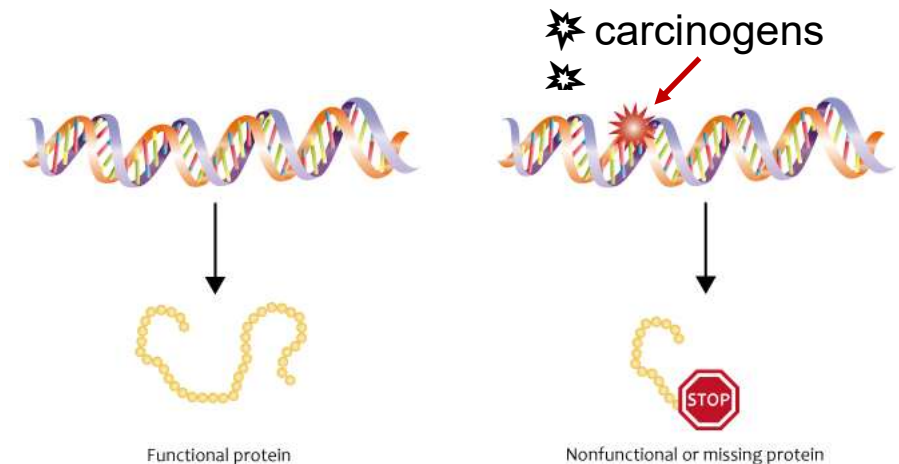
What happens when regulation goes wrong?

- *Remember:* genes (made of DNA) code for proteins
- *Remember:* mutations in genes can = nonfunctional proteins
- *Remember:* the checkpoints regulating the cell cycle are proteins

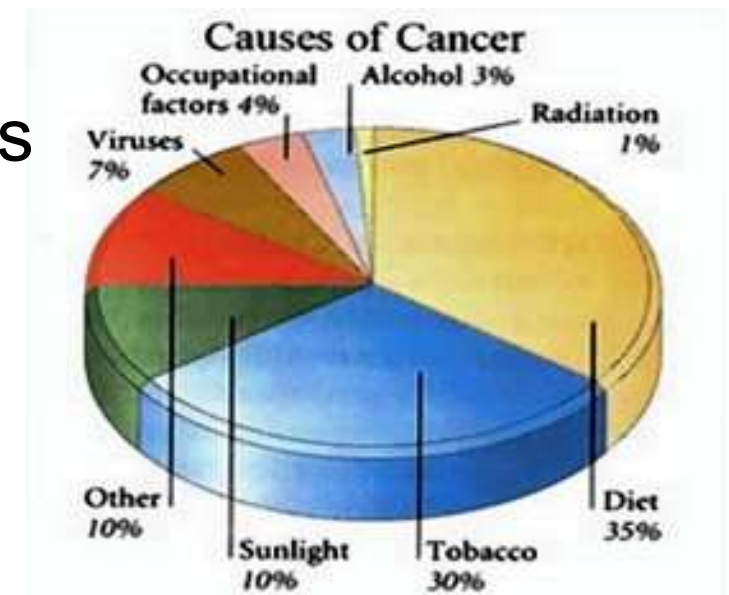
– So a mutation to the gene that codes for a checkpoint protein can = a nonfunctional checkpoint



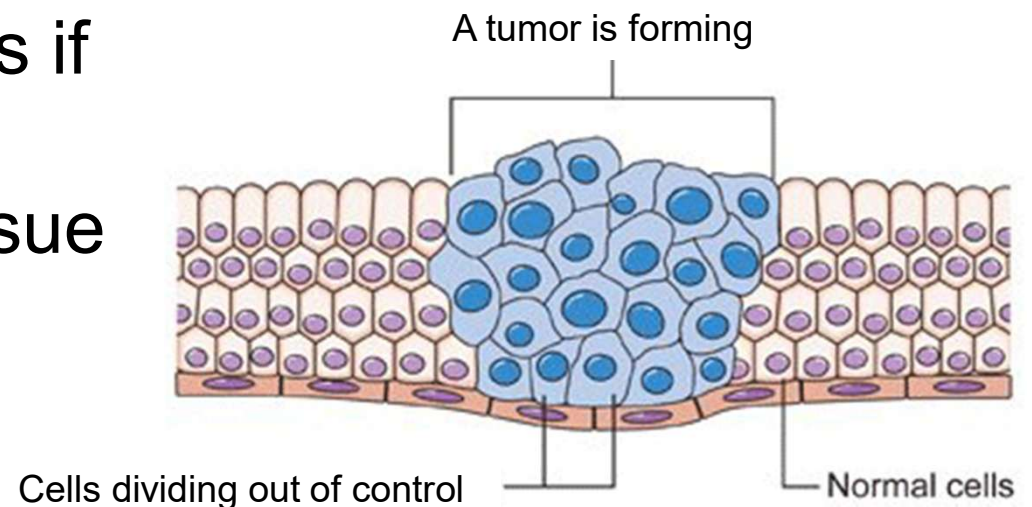
- If 1 checkpoint is not working, there are still 2 others
- But what if all 3 fail?
 - Seems unlikely that 1 cell would have multiple mutations at all 3 genes coding for the checkpoints: odds of winning the lottery
 - But: BILLIONS of cells are dividing in our bodies every day – that's a lot of lottery tickets!
 - plus, more & more **carcinogens** are in our environment, increasing the risk of mutations



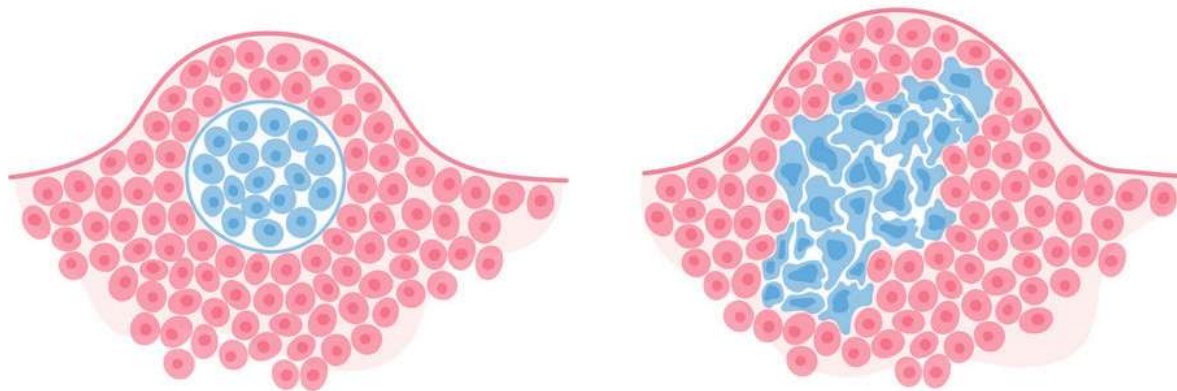
- Carcinogens are common in our environment:
 - Some **viruses** (HPV mutates DNA in cervical cells)
 - Some **chemicals** (asbestos, pesticides, etc.)
 - **Tobacco** (2nd leading cause of death in US)
 - **Radiation & sunlight** (x-rays, UV light)
 - **Animal proteins & fats**
 - e.g. hormones fed to animals linked to mutations in breast cells
 - e.g. normal red meat linked to mutations in colon cells



- So what happens if all of the checkpoints fail to show up & the cell cycle becomes unregulated?
 - Cells can grow out of control & form a **tumor**
 - There are 2 types of tumors:
 - **Benign tumor** = large mass of cells that is stationary
 - only dangerous if it damages the surrounding tissue



- **Malignant tumor** = cells dividing out of control that are able to break off & move around the body, invading & damaging other organs & starting new tumors – this movement is called **metastasis**
 - **Cancer** = diseases involving malignant tumor cells



Benign tumor

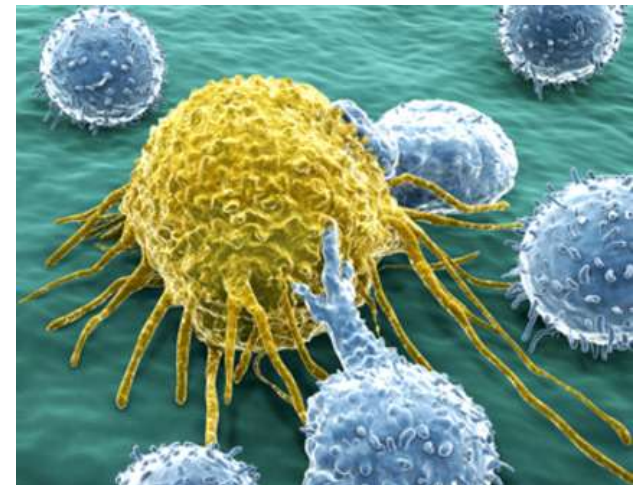
- Non-cancerous
- Capsulated
- Non-invasive
- Slow growing
- Do not metastasize (spread) to other parts of the body
- Cells are normal

Malignant tumor

- Cancerous
- Non-capsulated
- Fast growing
- Metastasize (spread) to other parts of the body
- Cells have large, dark nuclei; may have abnormal shape

- Cells with mutations form in our bodies all the time, especially since carcinogens are so common – so why haven't we all be diagnosed with cancer?
 - The immune system recognizes & destroys mutated & cancerous cells before they can spread
- Then why does anyone get cancer?
 - there are some cancerous cells that evade the immune system
 - the immune system may not be functioning at optimal levels
 - too many carcinogens may overwhelm the immune system

white blood cells attack a cancer cell



- The majority of cancers are caused by some factors that are under our control
 - e.g. exposing ourselves to carcinogens & not keeping our bodies in good health
 - *tobacco use, diet (fried foods, animal protein), sun exposure, environmental pollutants, infections, stress, & physical inactivity*
- In the media there tends to be more focus on *treatments* rather than *prevention*, but cancer is considered a “largely preventable disease”

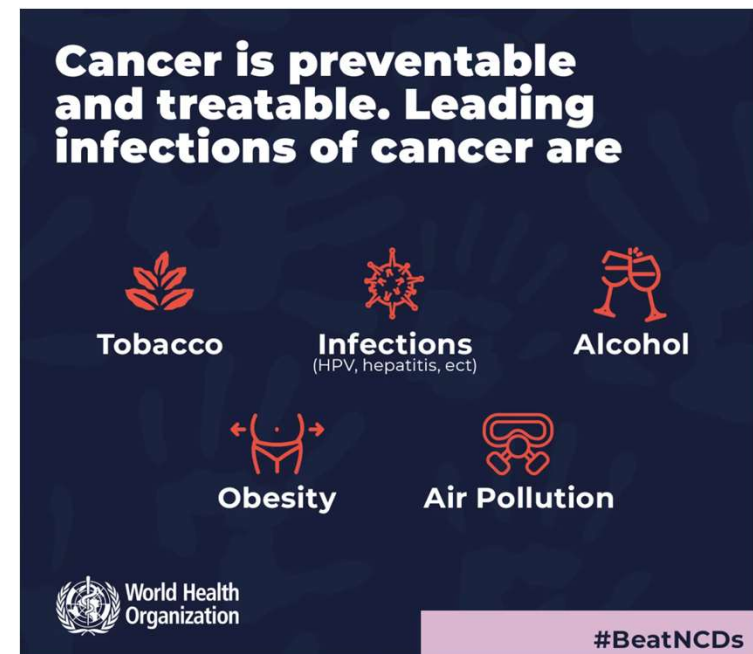
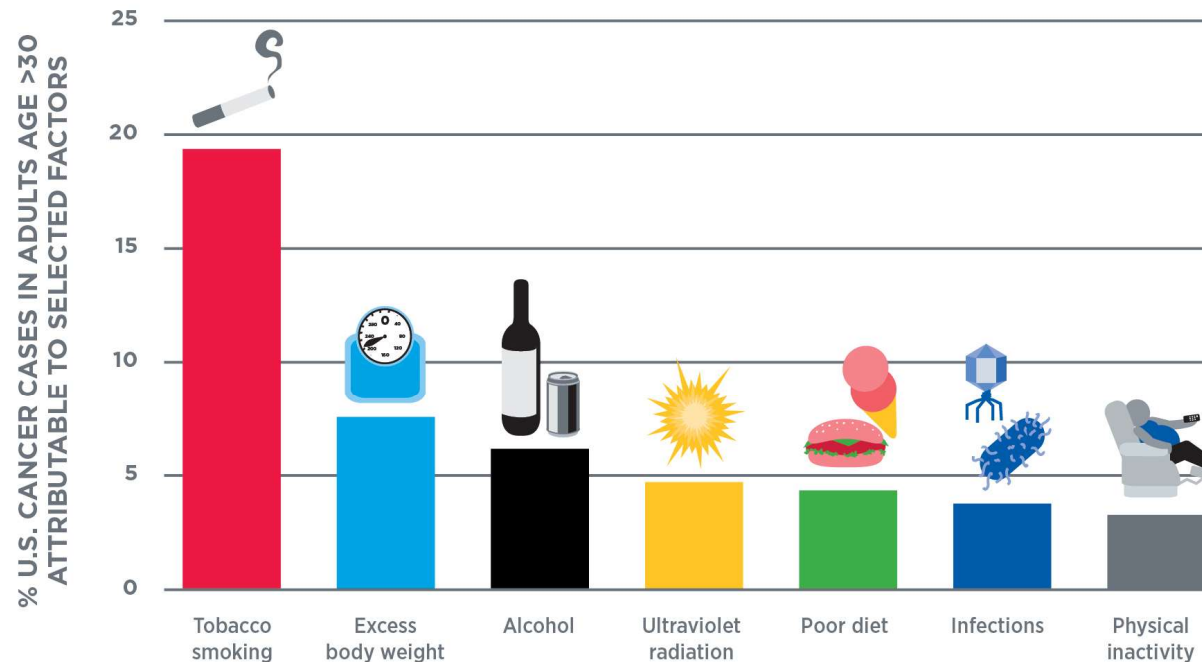


FIGURE 2**INCREASING CANCER RISK**

Research has identified numerous factors that increase an individual's risk for developing cancer. By modifying behavior, individuals can eliminate or reduce many of these risks and thereby reduce

their risk of cancer. Developing and implementing additional public education and policy initiatives could help further reduce the burden of cancers related to preventable cancer risk factors.

Data from (46). Figure adapted from (15)

American Association for Cancer Research (AACR) Cancer Progress Report 2019

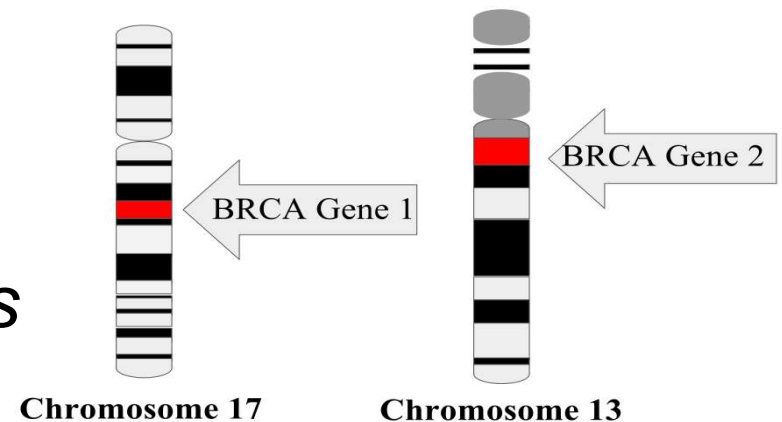
■ How do we prevent (or reduce our risk of) cancer?

- 1. Healthy diet & physical activity (*keeps immune system at peak*)
- 2. Avoid carcinogens as much as possible (*UV rays, tobacco, pesticides & other pollutants, etc.*)
- 3. Vaccinations (*HPV & Hepatitis B*)



American Association for Cancer Research (AACR) Cancer Progress Report 2019

- ~5% of all cancer has a genetic link
 - There are genes a person can inherit which alter the cell cycle, making it more likely that malignant cells will occur
 - e.g. BRCA1 & 2 are genes that produce proteins that help repair damaged DNA – a damaged gene in either location leads to increased risk of breast & ovarian cancer
 - *only a small percent of breast & ovarian cancer are due to these mutations – they are not common*



- What are the treatments for cancer?

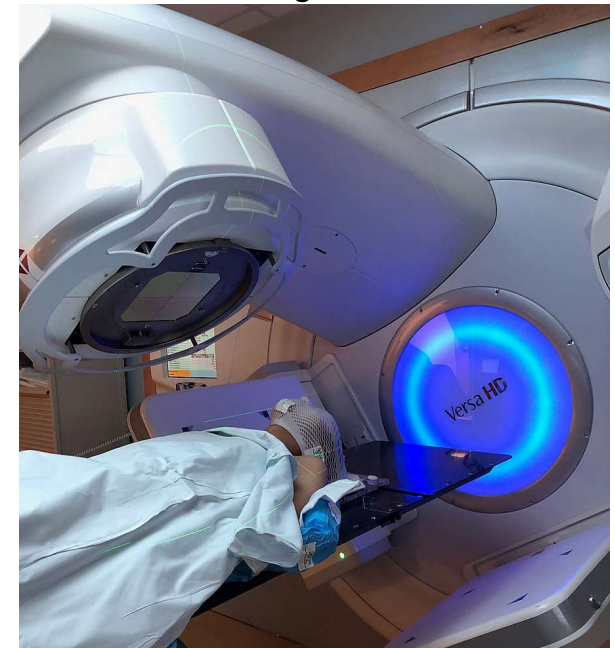
- Treating cancer is challenging: hard to target & kill some of our cells without killing the healthy ones

- **Surgery:** cutting out the harmful cells is always the 1st line of defense

- **Radiation:** if a malignant tumor hasn't spread (metastasized), radiation directed at the tumor kills the cells

- cannot be used on whole body (would kill all cells, even healthy ones)

A cancer patient receives a beam of radiation energy from this large machine



- What are the treatments for cancer?
 - **Chemotherapy:** drugs that stop cells from dividing (mitosis) – affects whole body so possible after metastasis
 - But these stop ALL cells in the body from dividing, which causes unpleasant side effects
 - e.g. no new hair follicle cells = loss of hair
 - e.g. no new intestinal cells = nausea
 - e.g. no new red blood cells = fatigue & shortness of breath

This woman is being treated with chemotherapy for breast cancer. Cooling mitts are placed on her hands and feet to ease deleterious effects on the nails. Similar strategies can be used to slow down hair loss.



- What are the treatments for cancer?
 - **Changes to diet & lifestyle:** can improve the bodies own immune response – can better fight cancer cells
 - Researchers are always searching for new options:
 - e.g. **Immunotherapy** = stimulating the immune system to recognize the cancer cells
 - e.g. **Targeted therapy** = attempts to develop drugs that ONLY target & kill cancer cells
 - e.g. **Angiogenesis inhibitors** = stop blood vessel growth – cuts off blood supply to tumor
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